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# WHEN DOES TRAINING ON DOWNSCALED IMAGES YIELD THE SAME GRADIENTS?

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## ABSTRACT

Diffusion transformers deliver strong image generation, but their training cost grows superlinearly with resolution. Recent work justifies training or sampling at reduced resolution on a spectral premise: at high noise, a downsampled latent preserves almost the full surviving signal. Whether a downsampled step also preserves the native *training gradient* signal, however, has remained unresolved. We explain how that signal changes under downscaling with a two-term account: a noise-dependent term governed by the downscale *ratio*, which decays at high noise as the spectral premise predicts, and a  $\sigma$ -independent floor governed by the target grid’s *absolute token count*, carried by the compute graph itself and removed by no noise level. The measured (route,  $\sigma$ ) map corroborates the account and uncovers structure the spectral picture cannot express: on the 1024→768 route, a window ( $0.65 < \sigma < 0.95$ ), predicted by no spectral criterion at any tolerance, where the downsampled gradient stays within a small margin of the native one. Training LoRA adapters with downsampled steps restricted to the routes and noise windows the map validates reduces training time by 14.6% at a fixed step budget while remaining near-native in weight space. Code is available at [https://github.com/sorryhyun/anima\\_lora](https://github.com/sorryhyun/anima_lora).

## 1 INTRODUCTION

The cost of a forward pass in a diffusion transformer (Peebles & Xie, 2023), the backbone of current text-to-image systems (Esser et al., 2024; Chen et al., 2024), grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. This cost structure makes the title question worth stating precisely: can the *gradient* of an adapter be computed on a downsampled latent as a direct substitute for the native one, with the model, the objective, and the  $\sigma$  distribution unchanged and only the spatial grid changed on some steps?

There is a well-studied reason to expect an affirmative answer *at high noise*. As the noise level  $\sigma$  grows, per-frequency signal-to-noise falls, and high spatial frequencies are the first to be masked by the noise, so that above a spectral crossover a downsampled latent carries *almost* the full surviving signal. Studies of this coupling, however, are concentrated on the *inference* side: scale-wise distillation (Starodubcev et al., 2025), spectral progressive diffusion (Xiao et al., 2026), and spectrally-guided noise schedules (Esteves & Makadia, 2026) run high-noise *sampling* steps at reduced resolution and judge the premise by the generated output. Where resolution is reduced during training (Jin et al., 2024; Karras et al., 2018; Chen et al., 2024; Hooeboom et al., 2023), the low-resolution stage is given its own, modified objective, and success is again judged from the final samples. By the spectral premise, substitution is safe *whenever the noise has masked the frequencies the coarse grid loses*.

This evidence, however, leaves the question open at *training* time: whether a downsampled step delivers the native gradient direction under an unchanged objective. Output-level success observes the substitution only through the confounds of a full training or sampling pipeline, and a small difference in a quality score does not imply a small difference in what a training step *learns*. What a training step consumes is the expected adapter gradient, and substituting a downsampled, re-encoded latent for the native one at training time perturbs more than the network’s input: the regression target carries the clean image at unit weight at every noise level, and the compute graph itself changes with the grid. We therefore decompose the gap between the downsampled and native gradients into two terms,

a noise-dependent term set by the downscale *ratio* and a  $\sigma$ -independent term set by the target grid’s *token count*, and measure each directly.

The contributions of this paper are as follows:

1. **Two accounts of the gap, and a debiased measurement that scores them** (§3, §4). We define the substitution gap at the gradient level and state two accounts of it in the same units. The first is the gap implied by the existing spectral argument, taken in its strongest tolerance-parameterized form (SPD, Xiao et al., 2026); it depends on the downscale *ratio* alone by construction. The second, the *gradient-perturbation* account, treats the substitution as a perturbation of both factors of  $g = J^T r$  and reduces, under mild, realistic assumptions, to a route amplitude on a universal mismatch curve *plus* a  $\sigma$ -independent term set by the target grid’s token count. The two differ structurally before any measurement (§3.2), and the resulting (route,  $\sigma$ ) map (§4.2), read with each route’s finite-sample bias estimated and subtracted and against a pre-specified margin, decides between them: it uncovers on the 1024→768 route a window ( $0.65 < \sigma < 0.95$ ) that the spectral account does not predict at any tolerance. Scored head to head on the same curves, the spectral account errs at 0.147–0.355 RMSE, whereas ours predicts held-out routes at  $\sim 0.07$ – $0.09$ , and the token-count law correctly predicts the  $\sigma$ -independent term on both held-out routes.
2. **Selective low-resolution training** (§5). Restricting the substitution to the window the map validates gives an opt-in  $\sigma$ -conditional trainer that reduces the forward-compute footprint by 15.1% and training time by 14.6% at a fixed step budget, with a weight-space endpoint within kernel-noise reach of a native retrain, rising to endpoint cosine 0.75 when the substitution is scheduled late at proportionally reduced saving (Table 1); a visual counterpart (Fig. 2) confirms the endpoint renders stay close to native.

## 2 RELATED WORK

**Flow matching and DiT.** Flow matching (Lipman et al., 2023) regresses a velocity field along prescribed probability paths between noise and data; rectified flow (Liu et al., 2023) takes the paths to be straight lines, giving the linear noising  $z_\sigma = (1 - \sigma)x + \sigma\epsilon$  and the constant target  $v = \epsilon - x$  used throughout, the training objective of current large text-to-image systems (Esser et al., 2024). The Diffusion Transformer (Peebles & Xie, 2023) replaces the U-Net denoiser with a transformer over patchified latent tokens, so a spatial grid enters the network only as a token sequence: token count grows quadratically with edge length and attention cost quadratically again in token count, the cost structure that makes resolution the expensive axis. Spatial position is now widely carried by a grid-dependent coordinate system, most commonly rotary embeddings (Su et al., 2024).

**Scale-wise and progressive-resolution diffusion.** Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogetboom et al., 2023) train at increasing resolution as a schedule, each low-resolution stage carrying its own stage-specific objective. A more recent family ties resolution to the noise level itself, on the spectral premise above, and has so far been studied on the *inference* side. Scale-wise distillation (SwD) (Starodubcev et al., 2025) states the claim directly (above the crossover, the noised downsampled latent is close in distribution to the downsampled noised latent) and samples early steps at reduced resolution. Spectral Progressive Diffusion (SPD) (Xiao et al., 2026) gives the argument its sharpest available form: a tolerance-parameterized per-frequency activation time under a diagonal Gaussian spectral model, a criterion on the per-band output of the *Bayes-optimal predictor* rather than merely on input SNR, from which an inference-time resolution schedule follows. Spectrally-guided noise schedules (Esteves & Makadia, 2026) apply the same reasoning to schedule design; where the premise does enter training, as in pyramidal flow matching (Jin et al., 2024), the objective is restaged per pyramid level rather than left native.

**Positional extension.** Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* that matches the downsampled grid’s relative phase geometry to native, in order to decompose the resolution-sensitivity floor, and adopt the  $\sigma$ -gated band schedule of Zhao et al. (2026) as a training-time refinement.

### 3 TWO ACCOUNTS OF WHAT DEMOTION COSTS

This section fixes the estimand (§3.1), then states two accounts of it in the same units: a training-gap interpretation built on the spectral account of the inference-side literature (§3.2), and the gradient-perturbation account (§§3.3–3.4). The two are not variants of one model: they disagree about which quantity governs a route’s cost, about what happens at pure noise, and about whether a single parameter can order all routes.

#### 3.1 THE ESTIMAND

**Objective and noising.** A flow-matching model regresses the constant-velocity transport between a clean latent and pure noise. With clean latent  $x$ , noise  $\epsilon \sim \mathcal{N}(0, I)$ , and noise level  $\sigma \in (0, 1]$ , the noised input is  $z_\sigma = (1 - \sigma)x + \sigma\epsilon$  and the target is the velocity  $v = \epsilon - x$ ; with caption condition  $c$  the loss is  $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$ .

**Demotion.** We write  $e_0 \rightarrow e$  for a *demotion*: the native latent at the route’s source edge  $e_0$  is replaced with a coarser-grid one at its target edge  $e$ . With  $y$  the image at its native resolution,  $\mathcal{E}$  the VAE encoder, and  $\mathcal{R}_e$  the pixel-space downscale to the tier- $e$  bucket (native aspect ratio preserved),  $x_{\text{src}} = \mathcal{E}(y) \rightarrow x_{\text{dem}} = \mathcal{E}(\mathcal{R}_e(y))$ , yielding a latent with proportionally fewer spatial tokens. The demoted step trains on  $z_\sigma = (1 - \sigma)x_{\text{dem}} + \sigma\epsilon$  with target  $v = \epsilon - x_{\text{dem}}$ , so the substitution changes both what the network sees and what it is asked to predict, but only through the spatial grid.

**The estimand.** For an arm  $a \in \{\text{src}, \text{reenc}, \text{dem}\}$ , where the source arm *src* trains on the native-resolution latent  $x_{\text{src}}$ , the demoted arm *dem* trains on  $x_{\text{dem}}$ , and the control arm *reenc* trains on the native latent decoded and re-encoded at native resolution, let  $\bar{g}_a(\sigma) = \mathbb{E}_\epsilon[\nabla_\theta \mathcal{L}]$  be the population per-bin mean adapter gradient of a query (image, caption). The *population demotion distance* of the route,  $d_{e_0 \rightarrow e}$ , is

$$d_{e_0 \rightarrow e}(\sigma) = 1 - \cos(\bar{g}_{\text{src}}(\sigma), \bar{g}_{\text{dem}}(\sigma)) = \frac{1}{2} \|\hat{g}_{\text{src}} - \hat{g}_{\text{dem}}\|^2, \quad (1)$$

with  $\hat{g}$  the unit-normalized gradients. Route-level scalars carry the *route* as a subscript, abbreviated to the target edge alone ( $d_e$ ) wherever the source is fixed by context; the control’s distance  $d_{\text{reenc}}$  carries its arm label instead, since it changes no grid (full convention: Appendix A.1). The *reported* quantity of every bin-resolved map and every fit in this paper is the *re-encoding excess*

$$\text{gap}_e(\sigma) := d_e(\sigma) - d_{\text{reenc}}(\sigma), \quad (2)$$

The control changes no grid, so its distance  $d_{\text{reenc}}$  prices the VAE round trip alone; subtracting it isolates the part of  $d_e$  attributable to the grid change.  $\text{gap}_e$  can be negative,  $d_e$  cannot. Every map in the body aggregates per example, taking the cosine per image and then the mean; the batch-aggregate object that SGD follows in batched training is a distinct estimand, treated in Appendix A.1.

#### 3.2 THE SPECTRAL ACCOUNT

The spectral argument of §2 supplies the field’s working model of the same gap; we now restate it in the units of Eq. 1. Its strongest available form is the tolerance-parameterized crossover of Xiao et al. (2026).

**Demotion deletes a band.** Write  $u^{(\omega)}$  for the frequency- $\omega$  component of a field  $u$  in the latent’s spatial Fourier decomposition, with  $\omega$  measured in cycles per sample of the *native* grid, so that the native Nyquist frequency sits at  $\frac{1}{2}$ . The downscale  $\mathcal{R}_e$  resamples the image with  $e/e_0$  times fewer samples per axis, and a grid at that density can represent no frequency above half its own sampling rate; the resampler’s anti-aliasing prefilter removes, rather than aliases, whatever lies above. Setting aside the VAE round trip, which the reenc control of §3.1 is constructed to absorb, demotion is an ideal low-pass:

$$x_{\text{dem}}^{(\omega)} = \begin{cases} x^{(\omega)}, & |\omega| < \omega_e, \\ 0, & |\omega| \geq \omega_e, \end{cases} \quad \omega_e = \frac{1}{2} \frac{e}{e_0}, \quad (3)$$

mapping every band at or above the coarse grid’s Nyquist frequency  $\omega_e$  to zero and acting as the identity on every band below it.

**Carried to the training gradient.** The noising  $z_\sigma = (1 - \sigma)x + \sigma\epsilon$  is linear and hence diagonal in the spatial Fourier basis, so it acts on each band independently. Modeling each clean-latent band as independently Gaussian,  $x^{(\omega)} \sim \mathcal{N}(0, P_\omega)$  with  $P_\omega$  the band’s clean-latent power, Xiao et al. (2026) attach to each band an activation time: a crossover  $t_\omega$  past which the Bayes-optimal per-band velocity prediction agrees, to within a tolerance  $\delta$ , with its data-independent limit  $\epsilon^{(\omega)}$ . The destroyed bands of Eq. 3 all sit at or above  $\omega_e$ ; on a decaying spectrum the lowest of them is the most powerful, so it crosses last, and the safe-substitution boundary for the whole route sits at  $t_{\omega_e}$ . The account attributes the route’s whole cost to those bands: whatever they contribute to the residual, and through the backward pass to the adapter gradient, perturbs  $\bar{g}_{\text{dem}}$  away from  $\bar{g}_{\text{src}}$ . The implied gradient gap is that contribution, gated at the Nyquist band’s activation time:

$$\text{gap}_e^{\text{spec}}(\sigma) = S_e(\sigma) \mathbf{1}[\sigma < t_{\omega_e}], \quad t_\omega = \frac{1}{1 + \sqrt{\delta / (P_\omega (1 + P_\omega - \delta))}}, \quad (4)$$

where  $S_e(\sigma)$  is the destroyed-band contribution to the distance of Eq. 1, and the tolerance  $\delta$  is the account’s single free parameter, instantiated in Appendix D.  $S_e$  can be computed. At the residual level the premise is complete: the Gaussian posterior mean is the classical per-band Wiener shrinkage (Wiener, 1949), so the cross-grid mean-residual mismatch is confined to the destroyed band, ordered across routes by destroyed-band energy, and follows in closed form from the power spectrum (Appendix A.3). Carrying it into the gradient units of Eq. 1 costs one calibrated gain, since the map passes through the network’s Jacobian, about which a spectral model of the *data* says nothing. Appendix D instantiates this pipeline when the account is scored.

Whatever the estimate of  $S_e$ , the account above commits to three limits, fixed by the *form* of Eq. 4 before any gradient is measured. **(a) One parameter orders every route:** all boundaries move together under the single tolerance, so the family is totally ordered. **(b) Ratio only:** the cut  $\omega_e = \frac{1}{2} e/e_0$  of Eq. 3 is a function of the downscale *ratio* alone, so the estimate is structurally blind to any absolute-size effect. **(c) No bias term:** for every route and every  $\delta$  the predicted gap vanishes above a finite boundary, so no  $\sigma$ -independent contribution is expressible.

### 3.3 THE GRADIENT-PERTURBATION ACCOUNT: WHAT DEMOTION PERTURBS

We start one level lower, from what the substitution touches in the gradient itself. Differentiating the loss of §3.1 gives, per draw and up to a constant,

$$g = \nabla_\theta \frac{1}{2} \|\hat{v}_\theta - v\|^2 = J^\top r, \quad r = \hat{v}_\theta - v, \quad J = \partial \hat{v}_\theta / \partial \theta,$$

with  $r$  the prediction residual and  $J$  the Jacobian through which the compute graph turns a residual into an adapter gradient. Demotion perturbs both factors: the residual, through the data the objective consumes, and the Jacobian, through the compute graph. The estimand is built from noise means, so split each factor into its noise mean and fluctuation under the averaging that defines  $\bar{g}_a$  in §3.1,  $r_a = \bar{r}_a + \tilde{r}_a$  and  $J_a = \bar{J}_a + \tilde{J}_a$ ; the arm’s mean gradient is then

$$\bar{g}_a(\sigma) = \mathbb{E}_\epsilon[J_a^\top r_a] = \bar{J}_a^\top \bar{r}_a(\sigma) + \mathbb{E}_\epsilon[\tilde{J}_a^\top \tilde{r}_a], \quad \bar{r}_a(\sigma) = \mathbb{E}_\epsilon[\hat{v}_\theta - v],$$

and the *mean residual*  $\bar{r}_a$  is a velocity-shaped field with a definite  $\sigma$ -curve for each arm.

Writing the demoted arm’s mean factors as source plus a perturbation,  $\bar{r}_{\text{dem}} = \bar{r} + \Delta\bar{r}(\sigma)$  and  $\bar{J}_{\text{dem}} = \bar{J} + \Delta\bar{J}$  (bare bars denote the source arm), subtracting the display across arms and expanding the product (Appendix A.2), the perturbation  $\delta g = \bar{g}_{\text{dem}} - \bar{g}_{\text{src}}$  decomposes as

$$\delta g = \underbrace{\bar{J}^\top \Delta\bar{r}(\sigma)}_{\text{data branch } B_e} + \underbrace{\Delta\bar{J}^\top \bar{r}}_{\text{graph branch } C_e} + \underbrace{\Delta\bar{J}^\top \Delta\bar{r} + \dots}_{\text{remainder } R_e}, \quad (5)$$

where the ellipsis collects the cross-arm difference of the noise-covariance couplings  $\mathbb{E}_\epsilon[\tilde{J}_a^\top \tilde{r}_a]$ .  $J_{\text{src}}$  and  $J_{\text{dem}}$  act on different grids, so the branches are not literal matrix products across arms; they are defined *operationally*, as the two independent perturbation directions in the shared adapter-parameter space where both gradients live. Read against Eq. 5, the spectral account of §3.2 keeps only the input-mediated part of the data branch, gated at Nyquist, and sets the rest to zero.

Two interventions follow immediately from the decomposition, rather than from experimental convention, and each isolates a branch. At  $\sigma = 1$  the input is pure noise in every arm, carrying no trace

of the image, so the input-mediated part of the data branch vanishes by construction; the regression target, however, carries the clean image at unit weight at every  $\sigma$ , so the mean residual retains an image-dependent survivor,  $\bar{r}_a(1) = x_a - \mathbb{E}[x \mid c]$  up to the frozen model’s approximation error (Appendix A.3): the endpoint reads the graph branch plus that target-mediated remnant of the data branch. Zeroing the image in input *and* target removes the survivor as well: the arms then differ in nothing but the grid, every data-mediated term vanishes by construction, and the x-zero probe reads the graph branch alone. The one question the two probes cannot settle, whether the survivor is *resolvable* in the angular estimand, is discharged in §4.3.

### 3.4 THE ANGULAR REDUCTION AND THE TWO-TERM ACCOUNT

It remains to read Eq. 5 in the units of the estimand. Split the perturbation  $\delta g$  into a rescaling of  $\bar{g}_{\text{src}}$  and a rotation away from it, with  $\kappa_{\parallel}$  and  $\kappa_{\perp}$  the two components in units of  $\|\bar{g}_{\text{src}}\|$ . The gap is then an exact saturating function of a single rotation-to-scale ratio,

$$d_e = 1 - \frac{1}{\sqrt{1 + \kappa_{\text{eff}}^2}}, \quad \kappa_{\text{eff}} = \frac{\kappa_{\perp}}{1 + \kappa_{\parallel}}, \quad (6)$$

so only the orthogonal component moves the estimand, and the cosine is *blind* to a parallel rescaling. Expanding at small perturbation and substituting Eq. 5 gives the *four-term angular expansion*: writing  $u^{\perp}$  for the component of  $u$  orthogonal to  $\bar{g}_{\text{src}}$ ,

$$d_e(\sigma) \approx \underbrace{\frac{\|B_e^{\perp}\|^2}{2 \|\bar{g}_{\text{src}}\|^2}}_{\text{data share } S_e} + \underbrace{\frac{\|C_e^{\perp}\|^2}{2 \|\bar{g}_{\text{src}}\|^2}}_{\text{graph share } \Phi_e} + \underbrace{\frac{\langle B_e^{\perp}, C_e^{\perp} \rangle}{\|\bar{g}_{\text{src}}\|^2}}_{\text{projected interaction } I_e} + R'_e, \quad (7)$$

with both shares non-negative, an interaction that can carry either sign, and a remainder  $R'_e$ ; the derivations and the remainder’s expanded form are discussed in Appendix A.4.

**From the expansion to a measurable form.** On the reported excess, Eq. 7 is exact but not yet an instrument: no term in it is separately measurable per bin. The excess itself does part of the work: the reenc arm changes no grid, so  $\Delta J_{\text{reenc}} = 0$ , its expansion degenerates to  $d_{\text{reenc}} \approx S_{\text{reenc}} + R'_{\text{reenc}}$ , and subtracting it cancels only the re-encode component of the data share, passing the graph share and interaction through untouched. The rest is carried by four assumptions, each tested by a designated probe in §4:

- **(i) Small remainder:**  $R'_e$  negligible over the claimed domain.
- **(ii) Negligible projected interaction:**  $|I_e|$  small against the retained shares.<sup>1</sup>
- **(iii) Graph-relative stationarity:**  $\|C_e^{\perp}\|/\|\bar{g}_{\text{src}}\|$   $\sigma$ -constant, since numerator and denominator share the residual’s scale, so that the graph share is flat in  $\sigma$ : a per-route constant, the *floor*.
- **(iv) Data-branch factorization:**  $\|B_e^{\perp}\| \approx a_e \|\Delta \bar{r}(\sigma)\|$ , a  $\sigma$ -independent route amplitude times the *norm* of the cross-grid mean-prediction-residual mismatch of Eq. 5, asserted to be route-independent and directly measurable as such (§4.3); the measured curve is an *empirical* closure of the mean residual’s exact posterior form (Appendix A.3), not an unconstrained fitted shape.

Under (i)–(iv) the expansion collapses to the *two-term account*,

$$\text{gap}_e(\sigma) \approx \underbrace{\frac{1}{2} a_e^2 \cdot (\|\Delta \bar{r}(\sigma)\| / \|\bar{g}_{\text{src}}(\sigma)\|)^2}_{\text{data term}} + \underbrace{\text{RoPE}_e + \text{Resid}_e}_{\text{graph term (the floor)}}, \quad (8)$$

a route amplitude  $a_e$  on a measurable mismatch curve plus a per-route constant; the supporting derivations are in Appendix A.4. The floor is written with two parts because a grid change touches

<sup>1</sup>Assumptions (i)–(ii) are not unconstrained: both discarded terms obey Cauchy–Schwarz bounds in the retained shares, the interaction by their geometric mean and the remainder entering at half power in its own share, so (ii) carries independent content only near share crossover (Appendix A.4).

two distinct things: the rotary coordinate system, in which every band of the position embedding accumulates phase at a different density across the image (RoPE<sub>e</sub>), and the non-positional graph statistics, attention softmax over a different token count and the coarse graph’s capacity to approximate the fine graph’s computation (Resid<sub>e</sub>).

The quadratic power of the data term is fixed by the cosine geometry *locally*; where the measured perturbation turns out large the licensed read is the un-expanded parent form, Eq. 6 evaluated at the same amplitude  $a_e \|\Delta\bar{F}(\sigma)\|/\|\bar{g}_{\text{src}}(\sigma)\|$  (recovering Eq. 8’s data term in the small-mismatch limit), whose data term *saturates* instead of overshooting. The ledger is as follows: Eqs. 6 and 7 are unconditional; Eq. 8 is conditional on the four assumptions; and the coefficients  $a_e$ ,  $\|\Delta\bar{F}(\sigma)\|$ ,  $\|\bar{g}_{\text{src}}(\sigma)\|$ , and each route’s floor are measured.

**The estimand forces a debiased instrument.** The angular reduction carries a metrological corollary. The estimand is a cosine of population means; any measurement replaces  $\bar{g}_a$  with a finite-draw average  $g_a = \bar{g}_a + \xi_a$ , and the draw noise  $\xi_a$ , being uncorrelated with the mean, enters the geometry of Eq. 7 as an orthogonal perturbation share of its own. Every finite-draw cosine is therefore attenuated below its population value, inflating the measured distance by a positive bias, and the bias is arm-dependent: per-draw gradient variance grows as the token count falls, so the inflation is larger on coarser grids and does not cancel in the excess of Eq. 2. The resulting artifact, a positive offset persisting at every  $\sigma$  and growing as the target grid shrinks, matches the signature of the graph floor. The same geometry dictates the remedy’s form. The draw noise is uncorrelated with the mean and independent across arms and draw sets, so it cancels in every expected cross term while inflating every squared norm: to first order the measured cosine factorizes,  $\cos(g_{\text{src}}, g_a) \approx \lambda_{\text{src}} \lambda_a \cos(\bar{g}_{\text{src}}, \bar{g}_a)$ , one attenuation factor  $\lambda_a = (1 + \mathbb{E}\|\xi_a\|^2/\|\bar{g}_a\|^2)^{-1/2}$  per arm. Each factor is measurable in place, since an arm’s cosine with an independent redraw of itself,  $\cos_{\text{self},a} = \cos(g_a, g'_a)$ , has expectation  $\lambda_a^2$ ; dividing by the geometric mean of the two self-cosines is Spearman’s classical correction for attenuation (Spearman, 1904),

$$\hat{c} = \frac{\cos(g_{\text{src}}, g_a)}{\sqrt{\cos_{\text{self},\text{src}} \cdot \cos_{\text{self},a}}}, \quad \widehat{\text{gap}} = 1 - \hat{c}, \quad (9)$$

which cancels the finite-draw attenuation of numerator and denominator to first order. The floor claim is therefore readable only through the debiased gap  $\widehat{\text{gap}}$ ; §4.1 instruments Eq. 9 and validates it.

## 4 EXPERIMENTS: BOTH ACCOUNTS, SCORED

This section scores both accounts of §3 against one measured object: debiased demotion-gap curves over the full  $\sigma$  axis, on routes neither account saw. We build the instrument (§4.1), read the per-route map its curves certify (§4.2), and then score both accounts against the same curves: the spectral account at the boundary and the curve level, the gradient-perturbation account term by term (§4.3). Each subsection states the prediction it discharges against criteria frozen before the runs.

### 4.1 THE INSTRUMENT

**Model and data.** All measurements use Anima (CircleStone Labs & Comfy Org, 2025), an open DiT-based flow-matching text-to-image model (2B parameters, 28 transformer blocks, 3D rotary position embeddings, and the Qwen-Image VAE, Wu et al., 2025), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length  $e \in \{512, 768, 896, 1024, 1280\}$ ; a tier- $e$  image occupies roughly  $(e/16)^2$  latent-patch tokens ( $\sim 4,100$  at 1024,  $\sim 3,000$  at 896,  $\sim 2,160$  at 768,  $\sim 1,000$  at 512). The probe adapter is a plain LoRA (Hu et al., 2022) checkpoint trained at native tiers: low-rank adapters are the dominant fine-tuning vehicle for models at this scale, and the trainer of §5 trains the same class.

**The gradient probe.** For each image and each of  $B$   $\sigma$ -bins we accumulate adapter gradients over  $D$  stratified noise draws per arm and compare arms by cosine similarity of the flattened accumulated gradients. The verdict run uses  $N = 40$  images and  $D = 12$  draws per bin, on a segmented grid of 14 bins in  $(0, 1)$ , dense below  $\sigma = 0.1$  and above 0.9, plus a  $\sigma=1$  endpoint bin, with deterministic kernels. The arms are: a **floor**, two independent draw sets at native resolution, the redraw null; the

**reenc** control of §3.1; one **demote** arm per edge  $e$ ; and, on debiased runs, per-arm **self-floors**, a second independent draw set for every arm. The  $\sigma=1$  endpoint bin is a distinct probe *mode*: its input is pure  $\epsilon$ , so the input-mediated data term vanishes by construction. An **x-zero** companion mode removes the image from input *and* target entirely. Both modes read only at the endpoint, since with the  $(1-\sigma)x$  term absent the lower bins are off-manifold (Appendix B), and together they carry the floor measurement of §4.3. The gap is in cosine units, a scale-free mismatch *fraction*, with bin-mean SEM 0.01–0.07 at  $N=40$ . Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with  $\|g\|$  across bins. Every bin-mean curve must pass a split-half reliability check over images, a discipline imposed after an earlier failure in which per-image rankings from the same gradients had reliability indistinguishable from zero.

**Finite-draw attenuation and the debiased estimator.** The arm-dependent attenuation of §3.4 is substantial at our operating point — the natural estimate of Eq. 1 also sets a redraw floor built from *two* finite-draw estimates against demote arms contributing one — and we observed it directly: a first-order variance estimate predicts spurious iso-direction endpoint gaps of  $\approx 0.02/0.05/0.15$  for 896/768/512, and the uncorrected 1024→896 endpoint gap indeed decays as  $+0.100/+0.035/-0.016$  at  $D = 4/8/16$ , a clean  $c/D$  decay (Appendix B). Two corrections, used together, remove it. First, *self-floors*: every arm runs a second independent draw set  $g'_a$ , restoring the floor’s two-estimate structure to every arm and supplying the per-arm self-cosines that Eq. 9 consumes; the floor is precisely the source arm’s self-cosine,  $\cos_{\text{floor}} = \cos_{\text{self,src}}$ , and the debiased gap  $\bar{g}$  is computed per image and bin. Second, *draw-count extrapolation*: the attenuation is the  $c/D$  decay just measured, so fitting  $\text{gap}(D) = \text{gap}_\infty + c/D$  over nested draw subsets recovers the draw-limit gap directly. The two are mutually checking, and the check passes: debiased fits come out  $D$ -flat ( $|c| \leq 0.05$  on the 512 route, against  $c \approx +0.29$  uncorrected), and the nested sweep extrapolates the native redraw floor to a self-cosine of 1.005 (bootstrap 95% CI [0.994, 1.016]). That is, the population per-bin native gradient is a *fixed direction*, and the entire redraw floor is finite-draw noise; remaining validation detail is in Appendix B.

**Margins and resolution.** A safety verdict needs two numbers, stated separately: a margin, the largest excess considered practically equivalent and a property of the question, and the instrument’s resolution, which sets only the power available to certify it. We fix the strict margin at  $\varepsilon = 0.02$ , comparable to the re-encode control’s own confidence half-width (Table 7); a verdict underpowered there is stated at the smallest margin its bounds do certify. A route is *certified safe at margin  $\varepsilon$*  in a bin where the one-sided non-inferiority bound clears it,

$$\text{UB}_{95}(\bar{g}) < \varepsilon, \quad (10)$$

and a *window* claim over several bins requires the simultaneous (Bonferroni-corrected) bounds to clear it in every bin. For a route whose true excess is zero the paired estimate  $\bar{g}$  fluctuates with standard error  $\text{SE}(N, D, \sigma\text{-bin})$ , so  $\varepsilon^* = 1.645 \text{ SE}$  is the *certification resolution*: a cell with  $\varepsilon^* > \varepsilon$  is *underpowered* at the margin, and a crossing there is reported but not counted as certification. At the verdict grid’s operating point of  $N=40$  and  $D=12$  per bin, the bin-level  $\varepsilon^*$  is 0.02–0.07, so window verdicts below are stated at the margins they do certify (0.09 for the windows of §4.2); endpoint-mode draw sweeps, with  $D$  up to 64, reach  $\varepsilon^* \approx 0.01\text{--}0.02$ , powered for the strict margin. All verdict maps report the *paired per-image excess*  $\text{gap}_{e,i} = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$ , trimmed of non-finite and  $|\text{gap}_{e,i}| > 1.5$  values, which cancels the shared per-image draw structure and stays stable at mid- $\sigma$  where the unpaired estimator is not; finite-draw cosines are compiler-kernel-path sensitive, so all pairings are computed within one run. Every verdict-carrying number in the main text is debiased; the iso-severity, tier-transfer, and positional-interpolation probes of §4.3 remain raw-estimator reads, flagged where cited (§6).

#### 4.2 THE MEASURED (ROUTE, $\sigma$ ) MAP

We measured the map directly: the measured (black) debiased gap curves of Fig. 1 carry the verdict, read per bin against the gray  $\pm\varepsilon^*$  band per Eq. 10 (all three routes on one axis, with the estimator context, in Appendix Fig. 8; per-bin numbers in Appendix Table 6); §4.3 scores both accounts on the same curves. Throughout, *safe* means the certification of Eq. 10 on the *per-example* object: the sense in which a demoted gradient is indistinguishable from a native one. Route by route: 1024→896 (floor 0.02–0.04) is safe on the window  $\sigma \in (0.5, 1)$ , certified at margin 0.09 simultaneously over the seven window bins; above  $\sigma = 0.94$  the bins are underpowered at margins below  $\approx 0.05$  and the

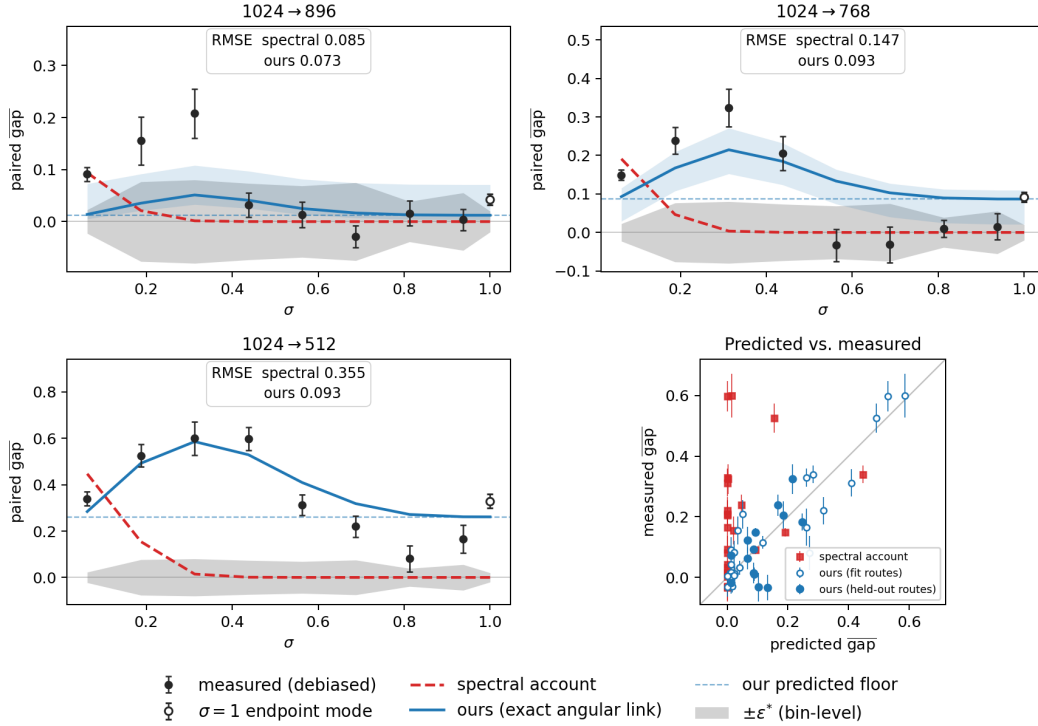


Figure 1: **The measured map, and both accounts on the same axes.** Per route, one set of measured debiased gap curves (Eq. 1, black; open markers detach the  $\sigma=1$  endpoint bin, a distinct probe mode) carries the verdict of §4.2, read per bin against the gray band, the bin-level  $\pm\epsilon^*$  certification resolution (§4.1). The same axes carry both predictions of §3: the spectral account (red dashed), transported through the residual→gradient bridge at the measured anchor tolerance, the whole family scoring alike (Appendix Fig. 6); and the gradient-perturbation account under the exact angular link (blue, Eq. 6; three-form comparison in Appendix Fig. 3), with each route’s predicted floor (blue dashed; dashed-to-solid is the data term) and a 95% full-pipeline bootstrap band ( $B=1000$ ; Appendix B).

$\sigma=1$  endpoint carries a small real gap (below), so the certified condition is  $\sigma \in (0.5, 0.94]$  (§6). 1024→768 and 896→768 (floors 0.06–0.09) certify no window at the strict margin, but 1024→768 retains a narrow low-excess window  $\sigma \in (0.65, 0.95)$ , with paired excess +0.02–0.04 per bin, certified at margin 0.09 simultaneously over its four window bins (0.072 pointwise). That excess sits *below* the route’s own floor, a dip the two-term account does not predict (the vector-resolved probe traces it to negative data–graph interference; §6); it is the window §5 stacks as a second route. Any route to 512 (floor  $\approx 0.30$ ) is *unsafe* in all fifteen bins, the lower bound clearing the margin in every one. Debiasing itself moved verdicts in both directions: roughly half the 768 route’s high- $\sigma$  plateau was estimator bias, softening “unsafe at every noise level” into the narrow window just stated, while the 896 endpoint shows a small *real* gap that single-estimate variance had buried ( $+0.034 \pm 0.011$  in the verdict grid’s endpoint bin;  $+0.019$  [ $+0.010, +0.030$ ] in the draw-limit sweep that §4.3 scores).

#### 4.3 BOTH ACCOUNTS, SCORED

Both accounts are scored against one measured object: the debiased gap curves of Fig. 1, read as §4.2 has just set out. The spectral account is scored first, at the boundary and then at the curve level; the gradient-perturbation account follows, term by term.

**The spectral account.** The spectral account’s prediction is monotone: the predicted gap is concentrated at low  $\sigma$  and vanishes at high  $\sigma$ , once noise drowns the destroyed band (equal-power crossovers  $\sigma \approx 0.14/0.15/0.20$  for 896/768/512, zero predicted gap above). The measurement disagrees in both regimes: the transported curves are near zero precisely where the measured curves peak (RMSE



0.147 at 768, 0.355 at 512), and above the crossover the floor persists (committed-region RMSE 0.027/0.049/0.218). Neither read depends on the tolerance or the residual-shape choice in the transport (within 0.01 RMSE; Table 5, Fig. 6, Appendix D).

**The data branch: a route-uniform residual mismatch.** Measuring the mean prediction residual  $\bar{r} = \mathbb{E}_\epsilon [\hat{v} - (\epsilon - x)]$  of §3.4 per grid and  $\sigma$  shows a strong, monotone  $\sigma$ -shape (cross-grid excess  $0.89 \rightarrow 0.36$  in relative  $L_2$  from  $\sigma=0.125$  to 1) that is the *same for every route* within  $\pm 0.02$ , across routes whose gradient floors span 0 to 0.3 (Appendix Table 11). This result refutes the residual-level prediction of §3.2: with no cross-frequency coupling, that mismatch is confined to the destroyed band and ordered by its energy, which differs substantially across these routes. Route-uniformity implies that the real mismatch is carried by cross-band structure the diagonal model cannot express, and that all route identity lives in  $J$ . The same contrast reads on assumption (iv): the measured route-uniformity is the shape factorization that assumption asserts, established here directly rather than assumed. The decay is smooth, Wiener-like shrinkage with no hard gate at the spectral crossover. How this monotone mismatch, read over the U-shaped total gradient norm, yields the measured mid- $\sigma$ -peaked curves is a post-hoc consistency read deferred to Appendix B.

**The graph branch: the floor exists.** At the  $\sigma = 1$  *endpoint* the input-mediated data term is zero by construction (§3.3); the debiased draw-limit endpoint floors are  $+0.019/+0.056/+0.304$  for 896/768/512 (Appendix Table 7), and the 512 value clears the pre-registered confirmation bar ( $\text{gap}_\infty \geq 0.15$ ) on 12 of 12 probe images individually. Three companion probes harden this number without adding to it. The  $x$ -zero probe reproduces the endpoint floors at every route (Table 7); a *target-strength sweep* (target  $\epsilon - \alpha x$ ,  $\alpha \in [0, 1]$ ) finds the paired debiased excess  $\alpha$ -flat, discharging the survivor question of §3.3 — the target-mediated gradient is real but demotion’s change to it lands along  $\hat{g}_{\text{src}}$ , to which the angular estimand is blind (Eq. 6); and a forward-only probe of the model’s caption-conditioned *prior* dissociates the prior from the floors’ route ordering, placing the ordering in  $J$ . The floor’s split into  $\text{RoPE}_e + \text{Resid}_e$  (Eq. 8) is then read by one origin-side intervention: evaluating the demoted grid at positionally-interpolated fractional rotary coordinates, which matches its relative phase geometry to native, *erases* the 768 endpoint floor and removes  $\sim 30\%$  of the 512 floor, the in-band 896 control untouched (a raw-estimator probe; §6). Re-anchored against the debiased floors,  $\text{Resid}_{768} \approx 0$ : what survives the coordinate fix is the harsh route’s  $\text{Resid}_{512} \approx 0.2$  bulk, a genuine non-positional graph residue, to which the capacity governor below attaches. (The erasure itself is no training lever; the frequency-selective banded alignment of §5 is the form that survives in-window.) Constructions, anchors, the parallel-landing decomposition behind the  $\alpha$ -flatness, and the floor ledger’s full numbers with their per-block depth localization are in Appendix B.

**The governors: ratio sets the amplitude, absolute size sets the floor.** With the floor isolated, what orders the routes? Two probes give a clean division of labor, and together they decide the equal-ratio disagreement, limit (b) of §3.2. *Absolute size, not ratio, sets the floor*: re-run one tier down, the route 896 $\rightarrow$ 768 fails its pre-registered bar with a high- $\sigma$  residual  $\sim 2\times$  the 1024 $\rightarrow$ 896 plateau despite a near-identical downscale ratio, while the two routes sharing the  $\sim 2,160$ -token target grid, 1024 $\rightarrow$ 768 and 896 $\rightarrow$ 768, land on the same floor despite different ratios (raw-estimator read, the separation confirmed by the debiased same-grid floors; Appendix Table 10). Near-equal ratio with a different target gives a different verdict; the same target with a different ratio gives the same floor. This refutes any pure ratio rule for the floor, and with it the spectral account’s only degree of freedom. Moreover, the floor grows monotonically as the *target* grid shrinks (0.02–0.04 / 0.06–0.09 /  $\approx 0.30$  at target  $\sim 3,000/2,160/1,000$  tokens), consistent with the reading that the floor measures how well the coarse graph approximates the fine graph’s computation. *Ratio, not absolute size, sets the amplitude*: a synthetic iso-severity route 1280 $\rightarrow$ 1120, with edge ratio exactly matched to 1024 $\rightarrow$ 896 at  $1.6\times$  its absolute target size, reproduces the 1024 $\rightarrow$ 896 curve bin-for-bin, crossover included (raw-estimator read; Appendix Table 9). Note what survives of the spectral account here: ratio is the right governor for the data term; its error is not its governor but its scope, mistaking one term for the whole gap.

## 5 $\sigma$ -CONDITIONAL TRAINING ON THE MEASURED-SAFE ROUTE

As an applied example, we introduce the measured-safe corpus route (1024 $\rightarrow$ 896) into a LoRA (Hu et al., 2022) trainer as an opt-in path, read off the map as measured, with one stated liberty: the gate is one-sided ( $\sigma > \sigma^*$ ), so demoted steps also land in the underpowered upper tail  $\sigma \in (0.94, 1]$

Table 1: Weight-space footprint:  $\cos(\Delta W_{\text{arm}}, \Delta W_{\text{native}})$  at matched training seed, mean  $\pm$  SD over three seed pairs, on two single-artist corpora (60/15 training images). Training uses deterministic kernels, including FlashAttention’s deterministic backward (Dao et al., 2022) (an identical retrain reads 1.000); the non-det. row retrains native at the same seed *without* deterministic kernels (one twin pair per corpus), so it reads run-to-run kernel nondeterminism alone. *late*: demotion only in the last 75% of steps; the stacked row adds the 768 low-excess window as a second route (1024 $\rightarrow$ 768 on  $0.65 < \sigma < 0.95$ , last 50%). **Bold**: closest endpoints.

| arm (vs. native, matched seed)     | corpus A                           | corpus B                           |
|------------------------------------|------------------------------------|------------------------------------|
| native, same-seed non-det. retrain | 0.264                              | 0.427                              |
| 896, $\sigma > 0.5$ , aligned      | $0.365 \pm .020$                   | $0.432 \pm .026$                   |
| + late                             | <b><math>0.753 \pm .019</math></b> | <b><math>0.770 \pm .009</math></b> |
| + late, stacked 768                | <b><math>0.753 \pm .017</math></b> | <b><math>0.771 \pm .009</math></b> |
| 896, every $\sigma$                | $0.183 \pm .009$                   | $0.236 \pm .009$                   |

Table 2: The proposed step, in full. Lines 3–5 (colored) are the entire change: the gate  $\sigma^*=0.5$  and the route are read off the measured map (§4.2), line 5 is the optional refinement, and lines 1–2 are only a re-ordering of a standard step, such that  $\sigma$  is drawn before the latent is fetched.

#### one LoRA training step

```

1  $\sigma \sim p_{\text{train}}$ 
2  $x \leftarrow$  native latent of  $y$ 
3 if  $\sigma > \sigma^*$  and  $y$  on-route:
4    $x \leftarrow$  demoted latent of  $y$ 
5   rotary  $\leftarrow$  banded( $\sigma$ )
6  $\epsilon \sim \mathcal{N}(0, I)$  at  $\text{shape}(x)$ 
7  $z \leftarrow (1 - \sigma)x + \sigma\epsilon$ 
8  $v \leftarrow \epsilon - x$ 
9 step on  $\|\hat{v}_\theta(z, \sigma, c) - v\|^2$ 

```

(§6). The intervention is deliberately small: three lines inside an otherwise standard flow-matching step, and no change to the objective, the optimizer, or the noise density (pseudocode in Table 2). We exercise it on a fixed-step grid: gated, late-scheduled (alone and with the 768 low-excess window stacked as a second route), and gate-removed arms  $\times$  3 training seeds  $\times$  two single-artist illustration corpora of contrasting style, one flat and graphic, one densely rendered (corpus A, 60 training images; corpus B, 15), 480 optimizer steps each, run with deterministic kernels (including FlashAttention’s deterministic backward, Dao et al., 2022) and paired RNG so every arm sees identical draws. At fixed steps the always-gated arm realizes a  $-14.6\%$  training-time saving over native training on our corpora; the late-scheduled arms demote only part of the run and save proportionally less.

**$\sigma$ -first draw.** The trainer draws each batch’s  $\sigma$  *before* fetching latents, from the unchanged training density. When every sample in the batch is above the gate ( $\sigma > 0.5$ ), the native 1024-tier latent is replaced with a cached demoted-grid counterpart produced by the probe’s own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream derives from the substituted latent. Off-route images always train native, as does validation.

**Banded rotary alignment,  $\sigma$ -gated.** Motivated by §4.3, a frequency-selective alignment is applied on demoted forwards only. It uses YaRN-style bands: a full positional-interpolation stretch for rotary bands with few rotations across the demoted extent, native spacing for high-frequency bands, a ramp between, and thresholds scheduled by a sigmoid gate in  $\sigma$  following Zhao et al. (2026). Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs, erasing the static variant’s low- $\sigma$  liability ( $+0.064 \rightarrow +0.033 \pm 0.025$ ) while preserving its high- $\sigma$  gains (paired  $-0.05$  vs plain demotion at  $\sigma \geq 0.59$ ).

**Weight-space endpoint footprint.** Table 1 compares the trained adapters directly in weight space, as the cosine between each arm’s endpoint  $\Delta W$  and its matched-seed native twin’s. Determinism makes the read exact: an identical retrain reproduces its twin at 1.000, and the same retrain with non-deterministic kernels reads 0.264/0.427 (corpus A/B), run-to-run kernel nondeterminism alone. Against that reference, demoting at *every*  $\sigma$  falls below the non-deterministic retrain (0.183/0.236), while demoting only above the gate ( $\sigma > 0.5$ ) matches or exceeds it (0.365/0.432). Late scheduling, demotion only in the last 75% of steps, brings the endpoint very close to the native weights (0.753/0.770), and, as the map’s low-excess read of the 768 window predicts, stacking the 1024 $\rightarrow$ 768 route onto the late rule leaves the cosine essentially unchanged (0.753/0.771). The visual counterpart (Fig. 2) confirms the read: matched (prompt, seed) renders from the late-scheduled arms are nearly identical to native.

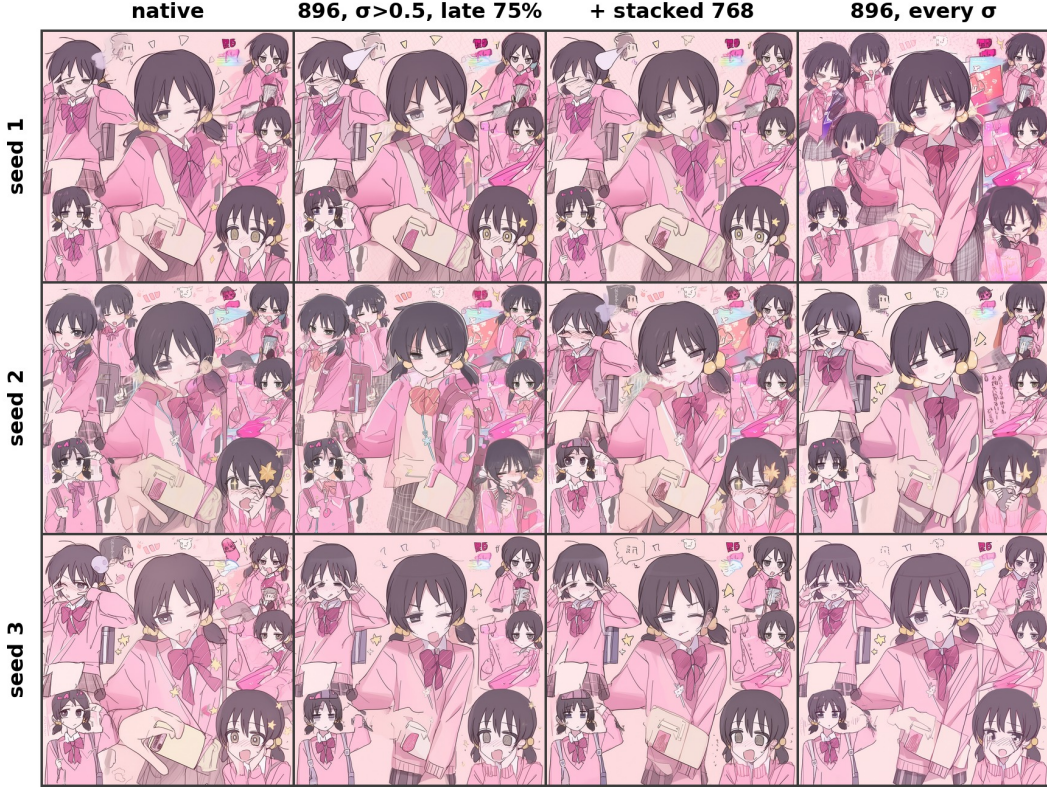


Figure 2: Visual counterpart to Table 1: matched (prompt, noise-seed) renders from the arms’ endpoints on a corpus-B display prompt, 20 steps; columns are four of the table’s arms (the unscheduled gated arm is omitted), rows the three training seeds. Moving along a row (arms, seed fixed) changes the render on the same visual order as moving down a column (training seed, arm fixed); the gate-removed arm (rightmost) drifts the most, matching its lowest cosine in Table 1.

## 6 LIMITATIONS

**One model family, one operating point.** All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint is the designated bound on how far the map generalizes, as is re-probing 1280→1024 after fine-tuning at the 1280 tier. The probe pool’s relationship to the adapter’s own fine-tune set was measured rather than conceded: a controlled 2×2 factorial (two LoRAs trained on opposite style clusters under frozen membership manifests, probed on both) replicates the map’s shape on both checkpoints with the adapter×probe-style interaction below instrument resolution, while the absolute redraw-floor level is checkpoint-dependent (Appendix E).

**Gradient-level, not end-task.** The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed by the exercise grid’s weight-space read at 480 steps rather than production scale.

**Account resolution and domain.** The account predicts held-out routes at ~0.07–0.09 RMSE, not within  $\varepsilon^*$  (the held-out protocol, its pre-registered gates, and the structural failures it exposes are quantified in Appendix C); its  $A(\text{ratio})$  governor rests on two distinct ratio values; the reduction’s domain excludes the 768 mid- $\sigma$  window, whose mechanism the vector-resolved probe has measured (negative interference) but which the two-term form still does not predict; and the floor law’s functional form is unidentified at our operating points. The graph-floor reading of the endpoint is also estimand-specific: the target-mediated perturbation is real at the vector level and lands parallel to the native gradient (Appendix B), so a magnitude-sensitive estimand, such as an un-normalized optimizer, could apportion the endpoint gap differently than the angular read does.

**Instrument resolution and coverage.** “Safe” is one-sided non-inferiority against a fixed margin ( $\varepsilon = 0.02$  strict; the map’s windows currently certify at 0.09 simultaneous,  $\approx 0.07$  pointwise), with the certification resolution ( $\varepsilon^* = 0.02\text{--}0.07$  per bin;  $\approx 0.01\text{--}0.02$  at the endpoint sweeps) setting the power available, not the margin. True gaps below  $\varepsilon^*$  are undetectable by construction,  $\sigma^*$  boundaries are localized only to a bin, and every failure to certify a window is  $(N, D)$ -relative: a larger campaign could certify (or refute) the 768 high- $\sigma$  window at the strict margin. The upper tail  $\sigma \in (0.94, 1)$  of the safe route is underpowered: its two non-endpoint bins resolve only margins above  $\approx 0.05$ , and the  $\sigma=1$  endpoint carries a small real gap ( $+0.034 \pm 0.011$ ); an upper-tail sweep at higher draw count is the designated probe to close the region, and until it lands the certified condition is  $\sigma \in (0.5, 0.94]$ . Verdict stability under the per-image trim rule of §4.1 has not been separately audited. The debiasing campaign covered the corpus verdict grid; the 1280-tier iso-severity, 896-tier transfer, depth-split, and PI-intervention probes remain pre-debiasing reads, and the 1280-tier curves enter the held-out validation on that basis.

## 7 CONCLUSION

When does training on downsampled images yield the same gradient direction? We answered by decomposing the demoted gradient’s perturbation into a ratio-governed data term and a  $\sigma$ -independent, token-count-governed graph floor, and by building the debiased estimator the question requires, since naive finite-draw estimation manufactures the very floor signature under test. Measured under that instrument, the spectral answer of the inference-side literature (safe once noise masks the frequencies the coarse grid loses) models only one part of one branch of what demotion perturbs: no tolerance reproduces the map, the mildest route’s crossover sits at  $3.5\times$  the equal-power prediction, and a  $2\times$  downscale certifies at no noise level, because the network function itself is grid-calibrated, partly through its rotary coordinate system and partly through an irreducible dependence of the computation on token count. The two-term account is predictive, not merely descriptive, reaching held-out routes at  $\sim 0.07\text{--}0.09$  RMSE against the spectral family’s  $0.147\text{--}0.355$ , and its map licenses a selective trainer: on our model,  $1024 \rightarrow 896$  at  $\sigma \in (0.5, 0.94]$  with a stackable  $1024 \rightarrow 768$  window, realizing  $-14.6\%$  training time at fixed steps with a weight-space endpoint within kernel-noise reach of a native retrain, rising to cosine 0.75 when demotion is late-scheduled at proportionally reduced saving. We offer the account and its instrument as the inexpensive, general test that any future “train part of the time at lower resolution” proposal should pass before it is believed.

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## A NOTATION AND DERIVATIONS

### A.1 NOTATION AND ESTIMAND FINE PRINT

**Arm labels versus route subscripts.** One notational convention holds throughout. Arm labels (src, reenc, dem) mark *whose* gradient, latent, or Jacobian an object is; the subscript on every route-level scalar is the *route*, written in full ( $d_{e_0 \rightarrow e}$ ) where several sources appear, and abbreviated to the target edge alone wherever the source edge is fixed by context, as it is in almost every measured statement; the abbreviation is an edge value, never an arm label, so it always instantiates numerically ( $d_e$ ,  $\Phi_{1120}$ ,  $\text{Resid}_{768}$ ). The control’s distance  $d_{\text{reenc}}$  (Eq. 1 with  $\bar{g}_{\text{reenc}}$  in place of  $\bar{g}_{\text{dem}}$ ) is the one exception that carries an arm label: the control changes no grid and has no target edge. Each table states which object it reports,  $d_e$  or the excess  $\text{gap}_e$ .

**The two aggregations.** In a batched training scenario Eq. 1 admits two aggregations over a probe set: per-example (cosine per image, then mean; the object every map in the body reports) or batch-aggregate (gradients summed over the batch before the cosine; the object SGD actually follows). These are different estimands, since the aggregation operator is part of the estimand, and no coefficient fitted on one is guaranteed to transfer to the other. At second order the two are related by a batch-size decomposition: writing each image’s demotion-induced gradient disagreement as a coherent mean  $b$  plus a zero-mean idiosyncratic deviation with covariance  $\Sigma_\eta$ , the batch-aggregate distance at batch size  $B$  is

$$\mathbb{E}[d_B] \approx \frac{\|P^\perp b\|^2}{2\|\mu\|^2} + \frac{\text{tr}(P^\perp \Sigma_\eta P^\perp)}{2B\|\mu\|^2},$$

with  $\mu$  the mean source gradient and  $P^\perp$  the projector off  $\hat{\mu}$ : an intercept (the coherent share, which never averages out) plus a  $1/B$  term. Monotone improvement with  $B$  is not automatic, since it would require an iid zero-mean disagreement model we have not verified, but where we have measured the batch object it is the more forgiving of the two: pooled-arm probes at pool size 4 collapse the 1024 $\rightarrow$ 896 excess to  $\approx 0$  at every bin  $\sigma \geq 0.625$  and the 1024 $\rightarrow$ 768 excess at  $\sigma \geq 0.875$ , the  $1/B$  term dominating there. What the collapse does not bound is the intercept, the coherent drift no batch size removes. A debiased verdict grid with pooled self-floors (pool 4,  $N=40$ ; paired arm–reenc gaps at  $B \in \{1, 4, 40\}$ ) bounds it: the  $a + b/B$  fit’s intercept is  $\leq 0.02$  (1024 $\rightarrow$ 896) and  $\leq 0.03$  (1024 $\rightarrow$ 768) at  $\sigma \geq 0.44$ , but persists over the lower bins (+0.04–0.09, +0.05–0.28, and +0.15–0.60 for 896/768/512) — aggregation confers safety only where the per-example map of §4.2 is already near-safe. The small real 896 endpoint gap (+0.042 in this run’s  $B=1$  read) averages out in the aggregate (+0.001).

### A.2 DERIVATION OF THE BRANCH DECOMPOSITION (EQ. 5)

Fix a query (image, caption), a  $\sigma$ , and an arm  $a$ . Per draw  $\epsilon$ , the adapter gradient of the  $\frac{1}{2}$ -scaled loss is the product  $g_a(\epsilon) = J_a(\epsilon)^\top r_a(\epsilon)$  of §3.3. Split each factor into its noise mean and fluctuation,  $J_a = \bar{J}_a + \tilde{J}_a$  and  $r_a = \bar{r}_a + \tilde{r}_a$  with  $\mathbb{E}_\epsilon[\bar{J}_a] = 0$  and  $\mathbb{E}_\epsilon[\tilde{r}_a] = 0$ ; the population mean gradient of arm  $a$  is then

$$\bar{g}_a = \mathbb{E}_\epsilon[J_a^\top r_a] = \bar{J}_a^\top \bar{r}_a + \mathbb{E}_\epsilon[\tilde{J}_a^\top \tilde{r}_a]. \quad (11)$$

Write the demoted arm’s mean factors as native plus a perturbation,  $\bar{J}_{\text{dem}} = \bar{J} + \Delta J$  and  $\bar{r}_{\text{dem}} = \bar{r} + \Delta \bar{r}$  (bars without an arm subscript denote the native arm). Subtracting Eq. 11 across arms and expanding the product,

$$\delta g = \bar{g}_{\text{dem}} - \bar{g}_{\text{src}} = \underbrace{\bar{J}^\top \Delta \bar{r}}_{\text{data branch } B_e} + \underbrace{\Delta J^\top \bar{r}}_{\text{graph branch } C_e} + \underbrace{\Delta J^\top \Delta \bar{r} + \mathbb{E}_\epsilon[\tilde{J}_{\text{dem}}^\top \tilde{r}_{\text{dem}}] - \mathbb{E}_\epsilon[\tilde{J}_{\text{src}}^\top \tilde{r}_{\text{src}}]}_{\text{remainder } R_e}. \quad (12)$$

This is an exact identity: no linearization is involved. “First-order” in the main text refers only to the content of  $R_e$ : it collects every term that is a product of two perturbations ( $\Delta J^\top \Delta \bar{r}$ ) or a difference of noise-covariance terms (the last two). Eq. 5 is Eq. 12 with the covariance difference abbreviated into its ellipsis.

One remark delimits the identity’s literal scope. When the two arms share a grid (as for the re-encoding control), every operation above is literal. Across grids,  $J_{\text{src}}$  and  $J_{\text{dem}}$  act on different token counts, so the differences  $\Delta J$  and  $\Delta \bar{r}$  require a fixed identification of the two output spaces (e.g.



spectral zero-padding of the coarse grid onto the fine grid’s bands); any such identification changes the split between  $B_e$ ,  $C_e$ , and  $R_e$  but not their sum  $\delta g$ , which lives in the shared adapter-parameter space regardless. This is why the main text defines the branches *operationally* (as the two independent perturbation directions in adapter-parameter space, isolated by the designated probes of §4.3) and treats Eq. 5 as fixing what each branch collects rather than as a computable matrix formula.

### A.3 THE MEAN RESIDUAL AS A POSTERIOR OPERATOR

This appendix derives the posterior form of the fixed-image mean residual, the exact split of the measured object into intrinsic posterior bias plus model approximation error, and the Gaussian closure referenced in §3.4. Fix a caption  $c$  and grid  $e$ , write  $a = 1 - \sigma$ , and let  $x \sim p_e(\cdot | c)$  be the grid- $e$  clean latent,  $z_\sigma = ax + \sigma\epsilon$  with  $\epsilon \sim \mathcal{N}(0, I)$ .

**Bayes field and fixed-image residual.** For the squared objective the population-optimal field is the conditional mean of the target  $v = \epsilon - x = (z_\sigma - x)/\sigma$ :

$$v_e^*(z, \sigma, c) = \mathbb{E}[v | z_\sigma = z, c] = \frac{z - m_e(z, c)}{\sigma}, \quad m_e(z, c) := \mathbb{E}[x | z_\sigma = z, c]. \quad (13)$$

Averaging over noise at fixed image ( $\mathbb{E}_\epsilon[\epsilon] = 0$ ,  $\mathbb{E}_\epsilon[z_\sigma] = ax$ ) gives the posterior form used in §3.4:

$$\bar{r}_e^*(\sigma; x) = \mathbb{E}_\epsilon[v_e^*(ax + \sigma\epsilon, \sigma, c) - (\epsilon - x)] = \frac{x - \mathbb{E}_{z_\sigma|x}[m_e(z_\sigma, c)]}{\sigma}. \quad (14)$$

With  $p_{\sigma,e}(z | c)$  the noised marginal, Tweedie’s identity for the linear interpolant reads  $m_e(z, c) = (z + \sigma^2 \nabla_z \log p_{\sigma,e}(z | c))/a$  for  $\sigma < 1$ , so equivalently

$$\bar{r}_e^*(\sigma; x) = -\frac{\sigma}{1 - \sigma} \mathbb{E}_{z_\sigma|x}[\nabla_z \log p_{\sigma,e}(z_\sigma | c)], \quad (15)$$

which makes the missing ingredient explicit: the numerical curve requires the grid- and caption-conditional smoothed score, i.e. a model of  $p_e(x | c)$ . At  $\sigma = 1$ , where  $z = \epsilon$  is independent of  $x$ ,  $m_e(z, c) \rightarrow \mathbb{E}[x | c]$  and  $\bar{r}_e^*(1; x) = x - \mathbb{E}[x | c]$ . With  $A_e$  the alignment operator that places the source-grid residual on the demoted grid (§4.1), the Bayes-level cross-grid mismatch is  $\Delta \bar{r}_e^*(\sigma; x) = \bar{r}_{\text{dem}}^*(\sigma; x_{\text{dem}}) - A_e \bar{r}_{\text{src}}^*(\sigma; x_{\text{src}})$ .

**What the instrument measures.** MSE optimality gives  $\mathbb{E}[v^* - v | z, c] = 0$  and hence a zero residual after a *joint* average over  $(x, \epsilon)$ ; it does not zero the fixed-image average  $\mathbb{E}_\epsilon[v^* - v | x, c]$ , which is Eq. 14 and generally nonzero. Writing the trained network as  $\hat{v}_{\theta,e} = v_e^* + q_{\theta,e}$ , the probe’s per-image object splits exactly as

$$\bar{r}_{\theta,e}(\sigma; x) = \bar{r}_e^*(\sigma; x) + \mathbb{E}_{z_\sigma|x}[q_{\theta,e}(z_\sigma, \sigma, c)] : \quad (16)$$

intrinsic posterior reconstruction bias, derivable in principle from  $p_e(x | c)$ , plus genuinely model-specific approximation error. The measured  $\|\Delta \bar{r}(\sigma)\|$  therefore has definite theoretical status, as an empirical closure of Eq. 14, but is not a pure model-error difference, and no claim in the paper requires it to be one: the two-term account consumes the measured curve as-is (assumption (iv)), and the trained-model residual is in any case the exact  $r$ -factor that the LoRA gradient  $g = J^\top r$  is built from.

**Gaussian closure.** The simplest nontrivial closure is a full Gaussian model  $x | c \sim \mathcal{N}(\mu_e, C_e)$  per grid. With  $Q_e(\sigma) = a^2 C_e + \sigma^2 I$ , the posterior mean is linear,  $m_e(z, c) = \mu_e + a C_e Q_e^{-1}(z - a \mu_e)$ , and substituting into Eq. 14 collapses (via  $I - a^2 C_e Q_e^{-1} = \sigma^2 Q_e^{-1}$ ) to

$$\bar{r}_e^*(\sigma; x) = \sigma Q_e(\sigma)^{-1}(x - \mu_e). \quad (17)$$

For paired source and demoted latents with covariance blocks  $C_{ss}, C_{dd}, C_{ds}$  and  $L_s = \sigma Q_s^{-1}$ ,  $L_d = \sigma Q_d^{-1}$ , the image-population mean squared mismatch is

$$\mathbb{E}[\|\Delta \bar{r}_e^*\|^2] = \text{tr}(L_d C_{dd} L_d^\top) + \text{tr}(A_e L_s C_{ss} L_s^\top A_e^\top) - 2 \text{tr}(L_d C_{ds} L_s^\top A_e^\top), \quad (18)$$

a  $\sigma$ -curve computable from paired *clean latents only*, with the source–demote cross-covariance  $C_{ds}$  load-bearing. A *diagonal Fourier*  $C_e$  makes Eq. 17 the per-band Wiener shrinkage of §3.2, confining the mismatch to the destroyed band and ordering it by destroyed-band energy: the residual-level



prediction that route-uniformity falsifies (§4.3). We tested whether any structured second-order closure suffices, entirely on stored paired clean latents (no denoiser forward): three nested covariance models (diagonal spectral; + within-band channel blocks; + octave-pair cross-band coupling), fitted on held-out-from-probe images and scored on the probe set against the measured curves. All three reproduce the curve’s monotone  $\sigma$ -shape (Pearson 0.94–0.97), its route-uniformity, the near-zero re-encoding control, and the  $\sigma=1$  endpoint, but over-predict the low- $\sigma$  amplitude by  $\sim 40\%$ , failing a pre-registered RMSE bar. Second-order data-only structure therefore recovers the shape but not the level, and the essential missing ingredients are the ones the identity names: caption-conditioning (the closure is caption-marginal), non-Gaussian posterior structure, and the  $q_\theta$  term of Eq. 16. The paper therefore retains the measured  $\|\Delta\bar{r}(\sigma)\|$  as its minimal honest closure. A theory–instrument comparison must also reproduce the instrument’s estimand, the split-half-corrected *relative*  $L_2$  distance of §4.1, i.e.  $\sqrt{\mathbb{E}\|\bar{r}_d - A_e\bar{r}_s\|^2}$  normalized by the mean of the two arms’ root-mean-square norms, rather than a raw mismatch norm.

#### A.4 THE ANGULAR REDUCTION: EXACT FORM AND FOUR-TERM EXPANSION

This appendix derives the two angular forms displayed in §3.4. Both follow from Eq. 1 and the split of  $\delta g$  by the native direction; no property of demotion enters. Write  $g = \bar{g}_{\text{src}}$ ,  $G = \|g\|$ ,  $\hat{g} = g/G$ , and  $u^\perp = u - (\hat{g}^\top u) \hat{g}$ ; decompose  $\delta g = G \kappa_\parallel \hat{g} + \delta g^\perp$  with  $\kappa_\parallel = \hat{g}^\top \delta g / G$  and  $\kappa_\perp = \|\delta g^\perp\| / G$ .

**The exact form.** The two components are orthogonal, so  $\|g + \delta g\|^2 = G^2(1 + \kappa_\parallel)^2 + G^2\kappa_\perp^2$  and

$$\cos(g, g + \delta g) = \frac{\hat{g}^\top (g + \delta g)}{\|g + \delta g\|} = \frac{1 + \kappa_\parallel}{\sqrt{(1 + \kappa_\parallel)^2 + \kappa_\perp^2}} = \frac{1}{\sqrt{1 + \kappa_{\text{eff}}^2}}, \quad \kappa_{\text{eff}} = \frac{\kappa_\perp}{1 + \kappa_\parallel}, \quad (19)$$

valid whenever  $1 + \kappa_\parallel > 0$  (the perturbation does not reverse the gradient direction). The gap of Eq. 1 is therefore exactly the saturating form displayed as Eq. 6 in §3.4. Since  $1 - \cos(\hat{g}_1, \hat{g}_2) = \frac{1}{2}\|\hat{g}_1 - \hat{g}_2\|^2$  for unit vectors, starting from either expression in Eq. 1 lands on the same result.

**The quadratic limit.** For small perturbations,  $1 - (1 + \kappa_{\text{eff}}^2)^{-1/2} = \frac{1}{2}\kappa_{\text{eff}}^2 + O(\kappa_{\text{eff}}^4)$  and  $\kappa_{\text{eff}}^2 = \kappa_\perp^2(1 - 2\kappa_\parallel + O(\kappa_\parallel^2))$ , so

$$d_e = \frac{\|\delta g^\perp\|^2}{2G^2} + \underbrace{O(\kappa_\perp^2\kappa_\parallel) + O(\kappa_\perp^4)}_{\text{beyond-quadratic}}. \quad (20)$$

**The four-term expansion.** Substituting  $\delta g = B_e + C_e + R_e$  (Eq. 12) into the leading term and expanding the square,

$$\|\delta g^\perp\|^2 = \|B_e^\perp\|^2 + \|C_e^\perp\|^2 + 2\langle B_e^\perp, C_e^\perp \rangle + \left(2\langle (B_e + C_e)^\perp, R_e^\perp \rangle + \|R_e^\perp\|^2\right),$$

whose first three terms, divided by  $2G^2$ , are the data share  $S_e$ , the graph share  $\Phi_e$ , and the projected interaction  $I_e$  of the four-term expansion, Eq. 7. The remainder  $R'_e$  has an exact expansion: the  $R_e$ -bearing terms above, the beyond-quadratic terms of Eq. 20, and the parallel-component correction, each suppressed by an extra power of perturbation size relative to the retained terms.

**From the four-term form to the excess.** The reported object is the excess over the control. The reenc arm changes no grid, so  $\Delta J_{\text{reenc}} = 0$ , and both control terms built from it vanish identically: its graph share  $\Phi_{\text{reenc}}$  and its projected interaction  $I_{\text{reenc}}$  each carry a factor  $(\Delta J_{\text{reenc}}^\top \bar{r})^\perp = 0$ . Writing Eq. 7 for each arm and subtracting term by term,

$$\begin{aligned} \text{gap}_e(\sigma) &= d_e(\sigma) - d_{\text{reenc}}(\sigma) \\ &\approx (S_e - S_{\text{reenc}}) + (\Phi_e - \Phi_{\text{reenc}}) + (I_e - I_{\text{reenc}}) + (R'_e - R'_{\text{reenc}}) \quad \text{Eq. 7, each arm} \\ &= (S_e - S_{\text{reenc}}) + \Phi_e + I_e + (R'_e - R'_{\text{reenc}}), \quad \Phi_{\text{reenc}} = I_{\text{reenc}} = 0 \end{aligned}$$

so the control collapses to  $d_{\text{reenc}} \approx S_{\text{reenc}} + R'_{\text{reenc}}$ , with  $S_{\text{reenc}}$  the re-encode share (measured  $\approx 0$  where probed: the control row of Table 7). The subtraction cancels only the re-encode component of the data share (up to a cross term suppressed by  $\sqrt{S_{\text{reenc}}}$ , bounded by the control’s near-zero reading), so  $\|\Delta\bar{r}\|$  is read net of re-encode; the graph share and the interaction pass through untouched, since the control has nothing to cancel them with.

**The drop bounds.** Both discarded terms obey exact bounds in the retained shares (Cauchy–Schwarz on the expansion above),

$$|I_e| \leq 2\sqrt{S_e \Phi_e}, \quad |R'_e| \leq 2\sqrt{(S_e + \Phi_e + I_e) \rho_e} + \rho_e + O(\kappa_\perp^2 \kappa_\parallel, \kappa_\perp^4), \quad (21)$$

with  $\rho_e = \|R_e^\perp\|^2 / (2\|\bar{g}_{\text{src}}\|^2)$  the remainder branch’s own share. The first says the interaction can never exceed the retained sum ( $2\sqrt{S_e \Phi_e} \leq S_e + \Phi_e$ ) and is automatically negligible wherever one share dominates the other, so assumption (ii) carries independent content only near share crossover, which is precisely where Appendix C finds it failing. The second says the first-order part of the remainder enters at half power, one factor of  $\sqrt{\rho_e}$  per retained share, so assumption (i) reduces to the branch-level statement that  $R_e$  is small against  $B_e$  and  $C_e$ , plus the small- $\kappa$  domain for the tail. The control counterpart  $R'_{\text{reenc}}$  needs no bound of its own: the control’s degenerate expansion is itself the measured near-zero row of Table 7, which pins  $R'_{\text{reenc}}$  jointly with  $S_{\text{reenc}}$ .

**What is not derived.** The loading  $\|\delta g^\perp\| = a_e \|\Delta \bar{r}(\sigma)\|$  (that the orthogonal gradient mismatch is proportional to the residual mismatch with a  $\sigma$ -independent route gain) is an empirical ingredient (the data-branch factorization at the amplitude level), fit and tested held-out in Appendix C. The geometry above fixes only how a given  $\delta g$  is read into gap units. What *is* derivable from first principles is only a one-sided envelope,  $\|\bar{J}^\top \Delta \bar{r}\| \leq \sigma_{\max}(\bar{J}) \|\Delta \bar{r}(\sigma)\|$ , which would turn the data term into an inequality in the observed mismatch norm given a  $\sigma$ -uniform bound on the mean Jacobian’s top singular value. Promoting the envelope to a proportionality with a  $\sigma$ -independent gain requires the mismatch *direction*  $\Delta \bar{r}(\sigma)$  to hold a  $\sigma$ -stationary alignment with  $\bar{J}$ ’s singular directions (isotropy, for instance, would give gain  $\|\bar{J}\|_F / \sqrt{n}$ ) while  $\bar{J}$  itself varies with  $\sigma$  through the network input; the image- and route-specific mismatch directions of Appendix B license no such axiom. Hence the loading stays an assumption with a designated probe rather than a theorem.

## B INSTRUMENT DETAILS AND SECONDARY READS

**Estimator.** Per image, arm, and  $\sigma$ -bin, adapter gradients are accumulated over  $D$  stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at  $D=8$  are not comparable *across* bins (the floor drops where  $\|g\|$  is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability  $\approx 0$  and is not used.

**Controls.** The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; in pre-debiasing runs its gap served as the validity band ( $\pm 0.04$ ; observed  $|\cdot| \leq 0.054$  across all bins of the main runs, per-module-group  $\leq 0.045$  in the split runs), a role superseded by the paired debiased read of §4.1. The redraw floor prices finite-draw noise. Density-weighting the uniform bins by the trainer’s  $\sigma$ -density reproduces earlier  $\sigma$ -marginalized measurements from the same instrument family (consistency check).

**Debiasing, implementation.** Under `--self_floor`, every arm (re-encoding and each demote variant) runs a second independent draw set from a disjoint seed stream, giving per-arm self-cosines alongside the native floor; Eq. 9 is then computed per image and bin, and the map read is the paired excess  $\text{gap}_{e,i} = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$  with non-finite values (negative self-cosines under the square root at low  $D$ ) and  $|\text{gap}_{e,i}| > 1.5$  trimmed (0–5 of 40 images per cell). Under `--draw_sweep`, draw prefixes are nested ( $D = 64$  contains the  $D \leq 32$  sets; prefix sums, no extra forwards) and the  $c/D$  fit of §4.1 is run on route means with a bootstrap CI over images. Finite-draw cosines are kernel-path sensitive: twin runs sharing a warm compiler kernel cache agree to  $|\Delta \cos| \leq 0.015$ , while a run compiled to a different kernel set lands up to  $\approx 0.3$  away at  $D=2$ , so gap/floor/self-floor pairings are not compared across processes, a guarantee the single-run instrument provides by construction; a `--deterministic` mode (bit-exact across twin runs) covers the paired training A/Bs.

**Finite-draw bias, measured.** Detail behind the magnitudes quoted in §4.1: the  $c/D$  fit to the uncorrected 1024→896 endpoint decay gives  $c \approx 0.5$ . The native floor makes the point sharpest: at the verdict grid’s  $D=8$ –16 the uncorrected endpoint self-cosine reads 0.79–0.85, a naive read pricing

the native gradient’s agreement with *itself* at a 0.15–0.2 gap, before the nested sweep ( $D = 4 \dots 64$ ) extrapolates it to the unit self-cosine reported in §4.1.

**The U-shaped denominator (behind §4.3).** The regimes of the total gradient norm  $\|\bar{g}_{\text{src}}(\sigma)\|$  ( $2.6 \rightarrow 0.4$  at  $\sigma \approx 0.3 \rightarrow 9.3$ ; Fig. 8b): large at high  $\sigma$ , where the model pulls composition out of the prior; small in the mid- $\sigma$  refinement regime; inflated again at low  $\sigma$  by irreducible  $\epsilon$ . A monotone absolute mismatch over this U-shaped norm yields the measured mid- $\sigma$ -peaked curves: bin-mean gap tracks  $1/\|\bar{g}_{\text{src}}\|$  (Spearman +0.55–+0.90). The renormalization check reads the amplification directly: multiplying each bin-mean gap by  $\|\bar{g}_{\text{src}}(\sigma)\|$  flips the anomalous low- $\sigma$  dip into the monotone-in- $\sigma$  maximum in every arm of both verdict runs, which is the low-norm amplification behind the mid- $\sigma$  peak of the measured curves. This shape read is post-hoc consistency analysis; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

**Endpoint and x-zero modes.** The  $\sigma=1$  endpoint bin feeds input exactly  $\epsilon$  (the input-mediated data term vanishes by construction). The x-zero mode zeroes  $x$  in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the  $(1-\sigma)x$  term absent, low- $\sigma$  bins are off-manifold and the  $\sigma=1$  read is primary. In that regime the residual is  $\approx -\hat{x}_{\text{prior}}$ , so “graph-dominated” includes the model’s grid-conditioned prior; the prior-distance probe (§4.3), a forward-only comparison of the model’s caption-conditioned prior across grids, subsequently dissociated the prior from the floor’s route ordering: its route distances are flat where the floors are strongly ordered, ruling out a resolution-conditioned prior or a training-distribution discontinuity as the carrier. In the target-strength ( $\alpha$ ) sweep the mid-sweep points  $\alpha \in [0.5, 0.75]$  are unreadable by construction and excluded: there the native residual  $\alpha x - \hat{x}$  passes near cancellation, the gradient norm dips ( $60 \rightarrow 33$ ) and the redraw floor falls to 0.87, so the estimator reads through a small gradient norm: the low-norm amplification above, surfacing along the  $\alpha$  axis. The well-conditioned anchors behind the  $\alpha$ -flatness claim of §4.3: 768 reads  $+0.070 \pm 0.015$  at  $\alpha=0$  vs  $+0.049 \pm 0.010$  at  $\alpha=1$  (control slope +0.003), and 512 reads  $+0.269 \pm 0.041$  vs  $+0.337 \pm 0.067$ .

**Parallel-landing decomposition of the target term (behind §4.3).** The  $\alpha$ -flatness has an exact mechanism. Because the objective is quadratic,  $\bar{g}$  is affine in  $\alpha$ , so the exact target-mediated gradient  $t = \bar{g}(1) - \bar{g}(0)$  is computable from the endpoint arms; measured, it is real and large ( $\|t_{\text{src}}\|/\|\bar{g}_{\text{src}}\| \approx 2.2$ , so  $J^T$  does not annihilate target content), but demotion’s change to it lands almost entirely *along*  $\hat{g}_{\text{src}}$  ( $\kappa_{\parallel} = -0.75/-1.18/-1.86$  against  $\kappa_{\perp} = 0.09/0.14/0.20$ , an 8–9 $\times$  parallel dominance reproducible across draw sets). The cosine gap is blind to a parallel rescaling (Eq. 6) and the orthogonal part enters only at second order ( $\kappa_{\perp}^2/2 \approx 0.004\text{--}0.02$ ), so the  $\alpha$ -flat result stands *because* the target term lands parallel, not because it vanishes. In this read the per-image orthogonal loading is  $\sim 10\times$  the aggregate’s: image-specific directions cancel in the mean, the same motif as the residual-direction read below. For 896 and 768 the floors sit at or below the verdict grid’s certification resolution, which the endpoint sweeps resolve only through their order-of-magnitude larger draw budget. The statement “the high- $\sigma$  plateau is the floor” therefore sharpens to: the floor is what the plateau converges to once the data term vanishes and the estimator can resolve it.

**Direction-resolved mismatch (behind §4.3).** The route-uniformity of  $\|\Delta\bar{f}(\sigma)\|$  is a property of the *amplitude* only. Saving the per-image mismatch vectors and comparing  $\Delta\bar{f}$  directions with a split-half attenuation correction, non-adjacent route pairs are near-orthogonal at low  $\sigma$  (corrected cosine 0.00–0.08 at  $\sigma \leq 0.375$ ) with only a weak shared component at high  $\sigma$  (+0.2–0.3 at  $\sigma \geq 0.875$ ); the top mode of a stacked SVD carries 0.33–0.36 of the energy against the 0.25 rank-4-uniform baseline (a rank-one common mode would give  $\approx 1$ ); and cross-image direction consistency is zero at every (route,  $\sigma$ ) cell (max +0.019 at split-half reliabilities 0.73–0.99), so the mismatch direction is fully image-specific, refuting a grid-conditional composition prior (“small canvas  $\Rightarrow$  portrait framing”) as the carrier, at least under full captions. What *is* shared is scale and spectral drift:  $\Delta\bar{f}$  energy migrates toward low frequencies as  $\sigma$  rises (low-third share 0.40  $\rightarrow$  0.67 by  $\sigma = 0.875$ ), composition-scale but per-image. Eq. 8 only ever consumes the norm, so the account is untouched; what sharpens is the open question, now “why is only the *amplitude* universal when the directions are image- and route-specific.” A closed-form candidate for cross-band coupling exists, namely the flow posterior covariance, whose off-diagonal  $D_z v^*$  entries are  $-((1-\sigma)/\sigma^3) \text{Cov}(x_{\omega}, x_{\omega'} | z)$  (Xing et al., 2026) and are set to zero by the diagonal-Gaussian premise; but its rank-one common-mode version is excluded by the SVD read above.

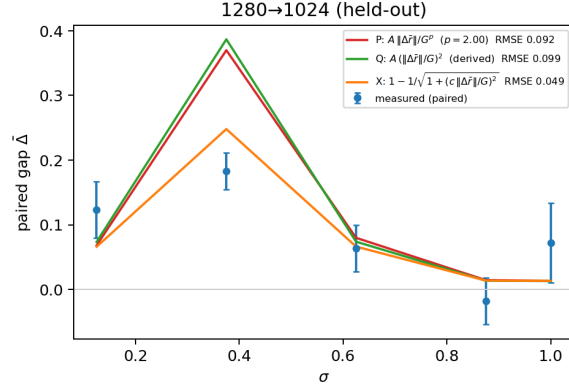


Figure 3: The second held-out route, 1280→1024, under all three functional forms for the data term of Appendix C (fitted power, derived quadratic, exact angular link; legend  $G := \|\bar{g}_{\text{src}}(\sigma)\|$ ), predicted from governors fitted on other routes. This is the one route where the forms visibly separate: the derived quadratic systematically overshoots the mid- $\sigma$  peak; the exact angular link’s saturation removes the overshoot (RMSE 0.049 vs  $\approx 0.095$ ), which is why the exact link is the form displayed in Fig. 1.

**Held-out validation and refit, sources.** The fit consumes the debiased paired curves of the 1024-tier verdict run, the 1280-tier curves (pre-debiasing paired reads; see §6), the route-uniform mismatch norm  $\|\Delta\bar{f}(\sigma)\|$  from the residual probe, and each run’s own bin-mean gradient norm  $\|\bar{g}_{\text{src}}(\sigma)\|$  (gap and norm always from the same process, per the kernel-path rule). The shared-power fit scans  $p \in [1, 2]$  jointly across fit routes with per-route weighted least squares; the exact-link fit is a per-route grid with profile-likelihood uncertainties on  $a_e$ . Oracle baselines are quadratic-in- $\sigma$  fits on the held-out data itself (the noise ceiling for a 3-dof curve); the spectral baseline on its committed region is gap = 0 for  $\sigma > \sigma_{\text{eq}}$ , and its curve-level transport is the bridge of Fig. 6.

**The floor law’s identifiability.** Two anchors and one held-out check cannot identify a functional form, and the cosine gap is the wrong scale to fit one in: it saturates (Eq. 6). Refitting the same anchors in the unsaturated perturbation-energy units  $\kappa_{\text{eff}}^2 = (1 - F)^{-2} - 1$  gives  $\tau \approx 860$  tokens and predicts  $F(2160) = +0.096$ , equally consistent with the measured  $+0.092 \pm 0.012$ ; the same two anchors pushed through  $e^{-\sqrt{n}/\ell}$  or a power law  $n^{-p}$  predict  $+0.085$  and  $+0.075$ . Every form falls within the union of the measured CI and the prediction’s own bootstrap spread, so the exponential is one member of a family our operating points cannot distinguish, and the mechanical reading of  $\tau$  ( $\approx 860$ –1040 tokens across the two unit conventions) as an effective-rank decay length of the early-block gradient operator is a hypothesis, not a law. It is a discriminable one: a spectral-tail account predicts a source-capacity dependence  $F \propto e^{-n/\tau} - e^{-n_0/\tau}$  where the absolute-target-capacity governor predicts none (at our operating points the secant correction sits within calibration slack, so the existing anchors do not discriminate); the designated run is a fixed-target, varied-source token ladder, not yet scheduled.

**Bootstrap bands and the leave-896-out lane (Fig. 1).** The bands are a full-pipeline image bootstrap ( $B=1000$ , fixed seed; 981 draws kept): resample images with replacement within each run ( $N=40 / N=24$ ), re-bin the paired curves, refit the exact-link ( $a_e, \Phi_e$ ) on every fit route, re-derive the ratio governor and floor law, re-predict.  $\|\Delta\bar{f}(\sigma)\|$  and the bin-mean  $\|\bar{g}_{\text{src}}(\sigma)\|$  are run-level instruments with no per-image decomposition and are held fixed. The replica includes the committed floor law’s positive-floor filter ( $\Phi_e > 0.005$ ): in 41% of draws the resampled 1120 floor clears the filter and the law re-anchors on the (896, 1120) legs rather than (512, 896). The bands therefore contain this branch variability, which is the reason §5 deploys the measured map rather than the predicted one. The 896 curve in Fig. 1 is the post-hoc leave-896-out lane under the exact link ( $c$  from the ratio-twin 1120 alone = 0.142 vs 0.150 fitted on 896 itself; floor law through 512+1120, whose fragile  $\Phi_{1120}$  leg enters the log fit clamped positive (36% of bootstrap draws clamp), giving  $F(3012) = +0.012$  vs measured  $+0.019 [+0.010, +0.030]$ ; curve RMSE 0.073).

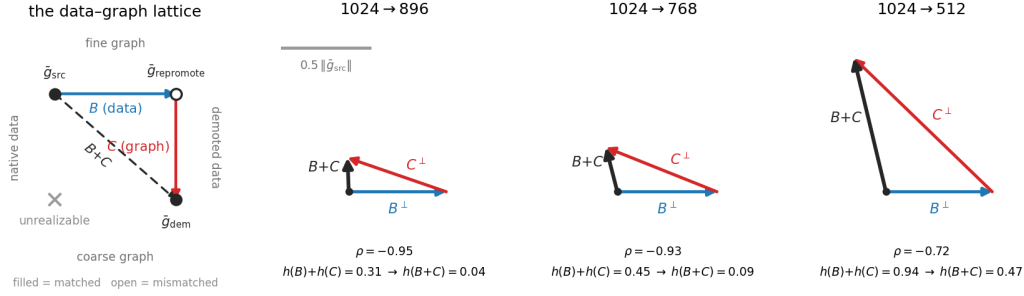


Figure 4: **The ledger’s geometry.** Left: the three arms as corners of the data–graph lattice.  $B$  and  $C$  are the partial differences along the lattice’s only realizable path — the fourth corner (native content on the coarse graph) does not exist — so the split is a round trip off the matched configuration ( $\bar{g}_{src}$ ,  $\bar{g}_{dem}$ ) through the single mismatched corner ( $\bar{g}_{repromote}$ ). Right: the measured legs tip-to-tail at  $\sigma = 0.563$  (Table 3), to a common scale in units of  $\|\bar{g}_{src}\|$ ; the drawn angle is the debiased  $\rho$ . Amplitude-matched legs leave only a small resultant (896, 768); the 512 graph leg is  $\sim 1.8\times$  the data leg, and the unmatched share survives as the realized gap.

**Interventional  $B/C$  ledger, implementation.** Routes  $1024 \rightarrow \{896, 768, 512\}$ ,  $\sigma \in [0.5, 1.0]$  in four bins plus the  $\sigma=1$  endpoint bin,  $D=8$  draws per bin,  $N=24$  images, deterministic mode, two independent draw sets per arm. Each arm’s draw-summed adapter gradient is stored per bin; the interventional split is  $B = \bar{g}_{repromote} - \bar{g}_{src}$  (the data intervention at fixed native graph: the demote–re-promote arm’s input passed through the downscale $\rightarrow$ upscale $\rightarrow$ encode pipeline but evaluated on the native grid) and  $C = \bar{g}_{demote} - \bar{g}_{repromote}$  (the graph intervention at fixed demoted data). The quadratic shares  $S$ ,  $F$ ,  $I$  and  $\rho = \cos(B^\perp, C^\perp)$  use cross-draw-set inner products to cancel shared draw noise; the debiased estimates are not confined to  $[-1, 1]$  (hence  $\rho = -1.24$  at one low-amplitude bin). The exact counterfactual angles  $h(B)$ ,  $h(C)$ ,  $h(B+C)$ , with  $h(X) = 1 - \cos(\bar{g}_{src}, \bar{g}_{src} + X)$ , are computed on the raw arm means; since  $\bar{g}_{dem} = \bar{g}_{src} + B + C$  by construction, the realized demotion distance is exactly  $h(B+C)$ , so each account’s predicted-to-realized ratio is direct. The no-interference scalar account  $S+F$  overpredicts the realized in-window gap  $2.4\text{--}3.8\times$  at 768 ( $1.6\text{--}5.8\times$  at 896,  $1.2\text{--}3.0\times$  at 512); the fully additive counterfactual  $h(B) + h(C)$  overpredicts  $3.5\text{--}8\times$ , the realized  $h(B+C)$  being only  $\sim 20\text{--}30\%$  of the additive sum in-window. Including  $I$  is what removes the overprediction; the vector account  $B+C$  is exact by construction. One caveat on the units of Table 3: in-window  $|B^\perp|, |C^\perp| \approx 0.5\text{--}1.0 \|\bar{g}_{src}\|$ , outside the quadratic truncation’s domain, so the truncated ledger  $S+F+I$  underpredicts realized magnitudes  $\sim 3\text{--}4\times$  (median ratio  $0.24\times$ ). The ledger licenses sign, decomposition, and window localization; gap magnitudes are quoted from  $h(\cdot)$ . Finally, the re-encoding proxy check re-references the data leg against the control arm,  $B_{reenc} = \bar{g}_{repromote} - \bar{g}_{reenc}$ :  $|B_{reenc}^\perp|$  agrees with  $|B^\perp|$  to  $\sim 4\%$  at the signal-carrying low- $\sigma$  bins (worst  $\pm 23\%$  only where  $B$  is already small), so the shared downscale $\rightarrow$ encode pipeline cost is a minor share of the data intervention. This closes the one open item on the floor ledger’s first share, where the re-encoding control (native decode $\rightarrow$ re-encode) is a *proxy* for the pipeline cost demotion actually pays.

**The ledger’s geometry: why the legs anti-align.** The near anti-parallelism of Table 3 is structural rather than incidental (Fig. 4). The three arm means are values of one map on a  $2\times 2$  data–graph lattice —  $\bar{g}_{src}$  at (native content, fine graph),  $\bar{g}_{repromote}$  at (demoted content, fine graph),  $\bar{g}_{dem}$  at (demoted content, coarse graph) — all living in the shared adapter-parameter space, so the differences are literal, and  $B$  and  $C$  are the partial differences along the path (native, fine)  $\rightarrow$  (demoted, fine)  $\rightarrow$  (demoted, coarse). That is the lattice’s only realizable path: its fourth corner, native content on the coarse graph, does not exist, since placing native content on the coarse grid is the demotion pipeline — the coarse graph forces the demoted data. The endpoints are the two content–graph *matched* corners (native content on its own grid; band-limited content on the grid whose Nyquist matches it); the unavoidable waypoint is the single mismatched corner. The split is therefore a round trip off the matched configuration and back —  $B$  creates the content–graph mismatch at fixed graph,  $C$  removes it at fixed data — so, to leading order in the mismatch, whatever share of each leg passes through it is the same vector with opposite sign, which fixes the sign of  $\rho$ . Its depth is then bookkeeping: projection off  $\bar{g}_{src}$  is linear, so  $|(B+C)^\perp|^2 = |B^\perp|^2 + |C^\perp|^2 + 2\rho |B^\perp||C^\perp|$ , and the ledger carries

two independent measurements — how far the waypoint sits from the corners (the leg amplitudes) and how nearly the endpoints coincide (the resultant) — with  $\rho$  their law-of-cosines consequence; the separately debiased sign column is a consistency check on the raw  $h(\cdot)$  reads, not a third fact. Two corollaries follow, and the data confirms both. Route verdicts reduce to amplitude matching, the realized  $h(B+C)$  collapsing where  $|B^\perp| \approx |C^\perp|$  (the window-center localization of Appendix C); and what no matching can cancel is the graph-intrinsic share of  $C$  with no data-side counterpart — the resultant collects exactly the residue the floor ledger below decomposes into  $\text{RoPE}_e + \text{Resid}_e$ , and at the  $\sigma = 1$  endpoint, where the input is  $\epsilon$  on every arm and the data leg has nothing left to act on, that uncanceled residue is the whole of the floor.

**Floor decomposition, detail (behind §4.3).** Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) localizes the floor in *depth*, not module type: early blocks (0–9, peaking at 3–8) carry  $\sim 3\times$  the late-block gap, uniformly across every module type within a block (at  $512/\sigma=1$  every type sits in 0.22–0.31; Table 12). The mechanism picture: the first  $\sim 10$  blocks build grid-calibrated token statistics, the divergence propagates into every parameter type’s gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Query and key projections show *zero* excess over value projections, which we initially read as refuting a rotary mechanism. That was an instructive error, since landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. Hence the origin-side positional-interpolation intervention (Table 13; the paired  $\Delta$ s are same-grid contrasts at matched draws, in which the finite-draw bias cancels to first order). The 896 route’s small floor sits below that probe’s paired resolution and remains undecomposed. A  $\sigma$ -resolved follow-up forecloses the route this suggests (training  $1024 \rightarrow 768$  through interpolated coordinates): with image content in the input the stretched forward is off-manifold, and the interpolated arm is *worse* than the plain demoted arm through  $\sigma \in [0.56, 0.81]$  (paired  $-0.05$  to  $-0.11$ ), winning only at  $\sigma \geq 0.94$ ; the 768 route’s data term is in any case fatal on its own ( $+0.09$ – $0.22$  over control across the window). On the non-positional residue, the attention-over- $N$  piece has inference-side prior art: attention-entropy corrections derive length-dependent temperatures to preserve the  $\log N$  entropy term (Li et al., 2025), and TIDE applies the same dilution correction jointly to resolution and diffusion timestep (Liu et al., 2026), the closest prior work to a  $\sigma$ -gated resolution correction, though on the inference side; our gate is training-side and set by measured gradient safety, not by an entropy invariance.

**The floor as a ledger.** Combining the designated probes of §4.3, the floor decomposes additively, each share measured by its own intervention:

$$\Phi_e = \underbrace{\text{reenc}}_{\approx 0 \text{ by control}} + \underbrace{\text{target content}}_{\text{lands } \parallel \hat{g}_{\text{src}}: \text{ angular share } \approx 0} + \underbrace{\text{RoPE}_e}_{\text{erased by PI at } \sigma=1} + \underbrace{\text{Resid}_e}_{\text{remainder}}$$

with totals 0.02–0.04 (896; below the PI probe’s paired resolution, undecomposed),  $\approx 0.07$  (768; RoPE the large majority, Resid  $\approx 0$ ), and  $\approx 0.30$  (512; RoPE  $\approx 0.10$ , Resid  $\approx 0.20$ ).

## C HEAD TO HEAD: PREDICTING HELD-OUT ROUTES

The probes of §4.3 test each account’s *structure*. The remaining test is predictive: fit the reduced account on three routes, then predict two held-out routes from the governors alone. Fit routes:  $\{1024 \rightarrow 896, 1024 \rightarrow 512, 1280 \rightarrow 1120\}$ ; held out:  $\{1024 \rightarrow 768, 1280 \rightarrow 1024\}$ . Per fit route the excess curve  $\text{gap}_e(\sigma)$  is fit with a route amplitude and floor on the measured route-uniform  $\|\Delta\bar{f}(\sigma)\|$  and each run’s own  $\|\bar{g}_{\text{src}}(\sigma)\|$ ; two governor models ( $A(\text{ratio})$  interpolating the fitted amplitudes, and an exponential floor law  $F(n) = F_0 e^{-n/\tau}$  in target tokens  $n$ ) then generate the held-out predictions with zero per-route freedom. Pass criteria were pre-registered in the analysis script before the numbers.

**Held-out results, and the comparison.** All four pre-registered gates pass (Table 4, Fig. 1). (i) Held-out  $1024 \rightarrow 768$ : RMSE 0.093, *better than an oracle quadratic fit on the held-out data itself* (0.105), and better than the spectral account both across all bins (0.147) and on the spectral account’s own committed region, where it predicts gap 0 (0.164 against 0.096 for ours). (ii) Held-out  $1280 \rightarrow 1024$ : RMSE 0.092 under the derived quadratic and 0.049 under the exact link, against

Table 3: The interventional  $B/C$  ledger behind Appendix C: per (route,  $\sigma$ -bin) orthogonal amplitudes (units of  $\|\bar{g}_{\text{src}}\|$ ), interaction geometry, the cross-set-debiased quadratic shares, and the exact counterfactual angles. In-window  $S+F+I$  is sign- and localization-accurate but underpredicts  $h(B+C)$  (small-perturbation truncation out of domain); magnitude claims in the text use  $h(\cdot)$ .

| route | $\sigma$ | $ B^\perp $ | $ C^\perp $ | $\rho$ | $S$   | $F$   | $I$    | $h(B)$ | $h(C)$ | $h(B+C)$ |
|-------|----------|-------------|-------------|--------|-------|-------|--------|--------|--------|----------|
| 896   | 0.563    | 0.55        | 0.59        | -0.95  | 0.103 | 0.115 | -0.206 | 0.107  | 0.206  | 0.037    |
|       | 0.688    | 0.42        | 0.39        | -0.91  | 0.036 | 0.059 | -0.085 | 0.083  | 0.077  | 0.062    |
|       | 0.813    | 0.20        | 0.23        | -0.94  | 0.016 | 0.019 | -0.033 | 0.022  | 0.027  | 0.010    |
|       | 0.938    | 0.10        | 0.10        | -1.24  | 0.002 | 0.003 | -0.006 | 0.007  | 0.006  | 0.003    |
|       | 1.000    | 0.05        | 0.06        | -0.71  | 0.001 | 0.001 | -0.001 | 0.002  | 0.003  | 0.002    |
| 768   | 0.563    | 0.57        | 0.68        | -0.93  | 0.139 | 0.200 | -0.309 | 0.107  | 0.346  | 0.088    |
|       | 0.688    | 0.56        | 0.57        | -0.93  | 0.100 | 0.136 | -0.218 | 0.106  | 0.242  | 0.098    |
|       | 0.813    | 0.35        | 0.42        | -0.93  | 0.049 | 0.067 | -0.105 | 0.059  | 0.110  | 0.035    |
|       | 0.938    | 0.14        | 0.17        | -1.02  | 0.005 | 0.009 | -0.014 | 0.010  | 0.033  | 0.008    |
|       | 1.000    | 0.07        | 0.09        | -0.62  | 0.001 | 0.004 | -0.003 | 0.003  | 0.014  | 0.013    |
| 512   | 0.563    | 0.60        | 1.09        | -0.72  | 0.166 | 0.403 | -0.372 | 0.085  | 0.854  | 0.466    |
|       | 0.688    | 0.63        | 0.88        | -0.86  | 0.153 | 0.332 | -0.388 | 0.107  | 0.623  | 0.305    |
|       | 0.813    | 0.57        | 0.72        | -0.87  | 0.152 | 0.226 | -0.323 | 0.131  | 0.285  | 0.126    |
|       | 0.938    | 0.25        | 0.29        | -0.96  | 0.026 | 0.036 | -0.058 | 0.028  | 0.201  | 0.037    |
|       | 1.000    | 0.13        | 0.15        | -0.88  | 0.007 | 0.010 | -0.015 | 0.007  | 0.379  | 0.098    |

Table 4: **Head to head** on the same measured curves: RMSE in cosine units over all  $\sigma$ -bins. The spectral column is the family transported through our bridge (Fig. 6) and is  $\delta$ -inert. Our two columns are the derived small-perturbation form (Eq. 8) and the exact angular link (Eq. 6), both with *zero per-route freedom* on held-out routes: amplitudes and floors come from the two governors of §4.3 fitted on  $\{1024 \rightarrow 896, 1024 \rightarrow 512, 1280 \rightarrow 1120\}$ . The oracle is a quadratic-in- $\sigma$  fit on the held-out data itself, the noise ceiling for a 3-dof curve, so beating it on  $1024 \rightarrow 768$  means the prediction is at the data’s own resolution.

| route                   | status    | spectral | two-term | exact link   | oracle |
|-------------------------|-----------|----------|----------|--------------|--------|
| 1024 $\rightarrow$ 768  | held out  | 0.147    | 0.093    | 0.093        | 0.105  |
| 1280 $\rightarrow$ 1024 | held out  | —        | 0.092    | <b>0.049</b> | 0.056  |
| 1024 $\rightarrow$ 512  | fit route | 0.355    | 0.093    | —            | —      |

oracle 0.056; inside the pre-registered  $2\times$  gate either way. (iii) The ratio governor holds at the amplitude level: the two ratio-0.875 routes agree at  $z = 0.14$  despite a  $1.6\times$  difference in target capacity. (iv) The floor law  $F(n) = 0.70 e^{-n/1041 \text{ tok}}$ , fit on the 512 and 896 floors, predicts  $F(4825) = +0.007$  for the 1120 grid (fitted  $+0.002$ ) and  $F(2160) = +0.088$  for the held-out 768 grid, against a measured endpoint floor of  $+0.092 \pm 0.012$ ; a post-hoc leave-896-out re-derivation from 512+1120 alone predicts the 896 floor at  $+0.019$  under the fitted-power form ( $+0.012$  under the exact link, Appendix B) against the measured  $+0.019$  [ $+0.010, +0.030$ ]. The floor law is thus correct on both floors outside its fit set, though its point estimates vary with the draw: a full-pipeline bootstrap puts  $F(2160)$  at 68% [ $+0.042, +0.091$ ], so the licensed read is agreement, not exactness.

The margin in Table 4 has a structural reading, not merely a numerical one. The spectral family’s error is not a bad tolerance, since  $\delta$  is inert, but a missing term and a missing governor: it has nothing to put in the high- $\sigma$  region where the floor lives, and nothing to distinguish 1280 $\rightarrow$ 1024 from a same-ratio route at a quarter of the token count. Our account’s held-out accuracy comes from precisely the two ingredients it adds.

**Which functional form, and what the fit does not identify.** Refitting the same pipeline under three forms for the data term (a fitted power  $A m/G^p$  with  $p$  shared and free, the derived quadratic, and the exact link) resolves the form question in the geometry’s favor twice over. The free exponent’s own optimum lands on  $p = 2.00$  exactly, recovering the derived quadratic on its own; and the exact link is the best held-out predictor (mean RMSE 0.071 vs 0.093–0.094), with its entire margin coming

Table 5: The spectral account (Eq. 4) instantiated on the measured latent spectrum ( $P$  at the 896/768/512 cuts = 0.025/0.029/0.065; RAPSD in Fig. 5): predicted safe boundaries  $t^*$  per route as the tolerance  $\delta$  sweeps its range.  $\delta_{\text{reenc}}$  is the measured re-encode noise floor, a zero-free-parameter anchor fixed by our own pipeline. No row reproduces the measured pattern (last line): the  $\delta$  that matches the safe route’s boundary predicts both failing routes safe by  $\sigma = 0.63$ , and no  $\delta$  can produce a floor.

| tolerance $\delta$                      | 1024→896      | 1024→768       | 1024→512          |                                         |
|-----------------------------------------|---------------|----------------|-------------------|-----------------------------------------|
| $(1 + P_\omega)/2$ (equal power)        | 0.14          | 0.15           | 0.20              | classical crossover                     |
| 0.025                                   | 0.50          | 0.52           | 0.62              | tuned to the measured 896 gate          |
| 0.01                                    | 0.61          | 0.63           | 0.72              | default of Xiao et al. (2026)           |
| $\delta_{\text{reenc}} \approx 10^{-5}$ | 0.98          | 0.98           | 0.99              | the parameter-free anchor               |
| <b>measured, debiased</b>               | $\approx 0.5$ | none certified | unsafe (8/9 bins) | floors .02–.04 / .06–.09 / $\approx .3$ |

from 1280→1024, where the quadratic systematically *overshoots* the mid- $\sigma$  peak and the exact link’s saturation removes the overshoot (Appendix Fig. 3). The governors are form-invariant. Two things the fit does *not* establish: the prediction is not within certification resolution anywhere ( $\chi^2/\text{bin}$  7.6–9.2 held-out under the quadratic), so the licensed claim is shape, magnitude class, and governors at  $\sim 0.07$ – $0.09$  RMSE; and the floor law’s functional form is unidentified at our operating points (the identifiability analysis is in Appendix B).

**The reduction’s domain: one signed failure, and its mechanism.** The 768 route’s mid- $\sigma$  window is the reduction’s one structural failure, and its shape is diagnostic. Measured excess sits at zero across  $\sigma \in [0.56, 0.94]$ , *below* the predicted  $\Phi_{768} \approx 0.09$  and outside the bootstrap 95% band (Fig. 1): a positive two-term form cannot produce a gap under its own floor, under any coefficients. In the four-term expansion (Eq. 7) this admits exactly two mechanisms, both pre-registered ahead of the probe: either the projected interaction is negative in the window ( $I_{768} < 0$ , in which case amplitude matching predicts the window center sits where the two perturbation legs have equal orthogonal magnitude, a testable localization), or the graph share is itself  $\sigma$ -dependent, rewriting assumption (iii) rather than (ii). The designated probe is vector-resolved: a demote–re-promote arm reads the shares per (route, bin) from the interventional split  $B = \bar{g}_{\text{repromote}} - \bar{g}_{\text{src}}$ ,  $C = \bar{g}_{\text{demote}} - \bar{g}_{\text{repromote}}$  (Appendix B, Table 3).

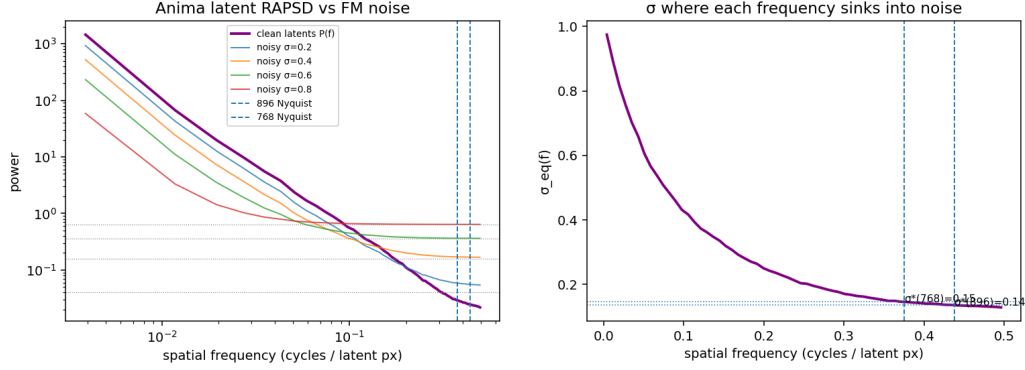
When the probe is run, the data select the first branch, with its localization confirmed:  $I_{768}(\sigma) < 0$  at every bin ( $-0.31$  at  $\sigma = 0.56$ , shrinking to  $-0.014$  by  $0.94$ ), with the two legs near anti-parallel in-window ( $\rho = \cos(B^\perp, C^\perp) \approx -0.93$ ), and the predicted window center lands at  $\sigma \approx 0.69$  ( $|B^\perp|/|C^\perp| = 0.98$  there), inside the measured window. Branch (ii) is ruled out:  $\Phi_{768}(\sigma)$  decreases monotonically to its endpoint value and never drops below it, so the endpoint reads of §4.3 are untouched. The anti-parallel geometry turns out to be universal ( $\rho \in [-1.24, -0.62]$  across every route and bin), as the round-trip reading of the split says it must be (Appendix B, *the ledger’s geometry*), so what distinguishes routes is amplitude matching, not sign; measured against the exact counterfactual angles, roughly three quarters of the additively-predicted gap is erased by the interference in-window (Appendix B). The window therefore *delimits the reduction’s domain*, and the derived four-term form represents the failure rather than excusing it.

## D THE SPECTRAL ACCOUNT: SPECTRUM AND INSTANTIATION

The tolerance  $\delta$  of Eq. 4 is the account’s one free parameter, and this appendix instantiates the whole family. The choice  $\delta = (1 + P_\omega)/2$  recovers the often-quoted equal-power (input-SNR) crossover  $\sigma_{\text{eq}} = \sqrt{P_\omega}/(1 + \sqrt{P_\omega})$  as one slice; Xiao et al. (2026) provide an inference-side default  $\delta = 0.01$ ; and our own pipeline fixes a zero-free-parameter anchor  $\delta_{\text{reenc}}$  by construction (below).

Table 5 evaluates Eq. 4 at the demoted grids’ Nyquist frequencies  $f_{\text{cut}} = \frac{1}{2} e/e_0$  on the  $N=40$  mean radially-averaged latent power spectrum (64 radial bins;  $P$  at the 896/768/512 cuts = 0.025/0.029/0.065; Fig. 5). Across  $\delta \in [0.001, 0.5]$  the mildest–harshes spread remains below 0.13, and the 1024→896 versus 896→768 boundaries (cut frequencies 0.4375 vs 0.4286) separate by no more than 0.01. The parameter-free anchor  $\delta_{\text{reenc}}$  of §4.3 is measured by the same estimator: RAPSD of the re-encode arm’s latent error (fresh resize–encode minus cached latent,





(a) Latent RAPSD against the flow-matching noise floor at four noise levels; dashed lines mark the demoted grids' Nyquist cuts. (b) The equal-power slice  $\sigma_{eq}(f)$  of the spectral family. Predicted  $\sigma^*$ : 0.136 / 0.146 /  $\sim 0.20$  for demotion to 896 / 768 / 512.

Figure 5: The spectral account, computed from the measured RAPSD and transported to the training question (§3.2): safety would be predicted from  $\sigma \approx 0.14$ . Against the measurement (Fig. 8) no tolerance reproduces the pattern (Table 5): the equal-power crossover is off by  $\sim 3.5\times$ , and the  $2\times$  downscale, predicted safe by every tolerance, certifies at no noise level.

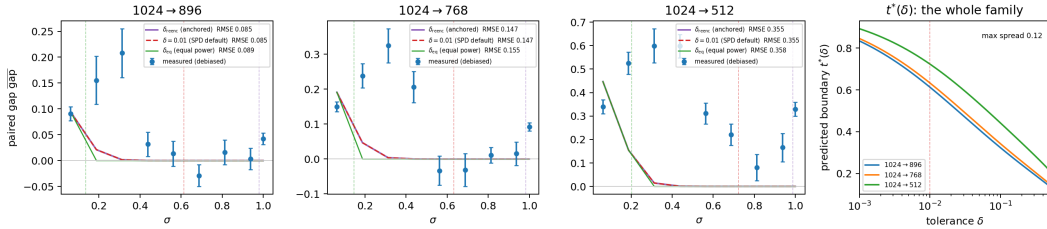


Figure 6: The *whole* tolerance family against the measurement: the detail behind the single red curve of Fig. 1. Left three panels: the measured debiased gap curves against the spectral account transported through our bridge (the diagonal model's destroyed-band mean-residual mismatch on the measured spectrum, through the measured  $\|\tilde{g}_{src}(\sigma)\|$  with a single gain calibrated on the one safe route and no floor, which the family cannot express), gated at  $t^*(\delta)$  (dashed verticals) for the equal-power slice, SPD's default  $\delta = 0.01$ , and the measured anchor  $\delta_{reenc}$ . The transported curve decays to zero by  $\sigma \approx 0.35$  on every route, so the two larger gates (0.61–0.72 and 0.98–0.99) act on an already-zero curve and coincide exactly; the equal-power gate (0.14–0.20) is the only one that truncates a nonzero curve, and it moves RMSE by at most 0.008. All three therefore land within 0.01 of each other:  $\delta$  is inert at the curve level. Right: the boundary family  $t^*(\delta)$ ; maximum route spread 0.125 in  $\sigma$  over  $\delta \in [10^{-3}, 0.5]$ .

the gradient probe's own control chain) on the same probe set, interpolated at each route's cut frequency. Measured ( $N=40, 64$  radial bins): error variance  $2.7 \times 10^{-5}$  against latent variance 0.419;  $D(f_{cut}) \approx 1.0 \times 10^{-5}$  at all three cuts (p10–p90 over images within  $[1, 2] \times 10^{-5}$ ; per-band SNR  $P(f_{cut})/D(f_{cut}) \approx 2.5\text{--}4.4 \times 10^3$ ), giving  $t^*_{reenc} = 0.98/0.98/0.99$  for 896/768/512.

**The “wrong  $\delta$ ” objection, closed by construction.** One might object that the right  $\delta$  has simply not been chosen: SPD's is an inference-side value selected on a speed–quality Pareto ablation over generated images (Xiao et al., 2026), and nothing ties it to gradients. Our setting fixes one *by construction*. The demotion recipe pays a resize–VAE–encode round trip whose latent error has measurable per-band power  $D(f)$ , expressed in Eq. 4's own units, so  $\delta_{reenc}(e) := D(f_{cut}(e))$  anchors the family with zero free parameters. Measured (above), that anchor is tiny and image-generic, and the boundary moves it forces are measured in Table 5 and Fig. 6: every boundary is pushed to  $t^* \approx 0.98$ , maximally conservative, and still structurally wrong in both directions at once: it moves all three boundaries together (spread  $< 0.01$ ), certifies the measured-safe route only above  $\sigma \approx 0.98$  where

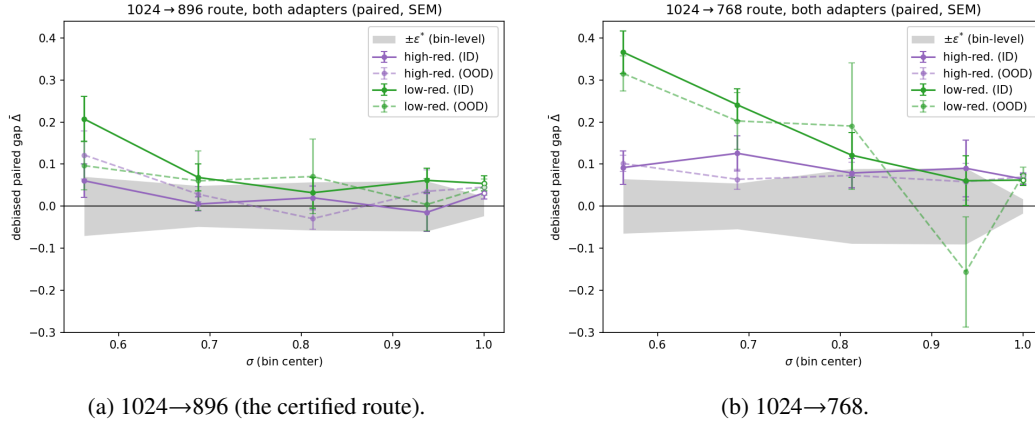


Figure 7: The Fig. 8a recipe repeated on two controlled adapters trained on opposite style clusters, one panel per route ( $N=48$  stems per adapter,  $D=8/\text{bin}$ , high- $\sigma$  window; same paired debiased estimator and trim; open markers the  $\sigma=1$  endpoint bin; axes shared across panels). Unlike the main-text figures, color encodes the *adapter* (high- vs low-redundancy cluster); solid curves are in-distribution stems, the adapter’s own style cluster ( $n=36$ : trained-on, held-out, and unseen-artist cells), and dashed curves are out-of-distribution stems from the opposite cluster ( $n=12$ ). Gray band: bin-level  $\pm\epsilon^*$  per Eq. 10, median of the two runs’ full- $N$  SEMs for the panel’s route. The map replicates: on the certified route all four adapter $\times$ distribution curves reach the band inside the window, on 768 all four sit above it until the top, and within each panel the dashed OOD curves track their solid ID counterparts.

the measurement certifies it from 0.5, and, like every member of the family, predicts the  $2\times$  down-scale eventually safe when no noise level is. The confrontation of §4.3 therefore does not hinge on the choice of tolerance: the structural failures of §3.2(a–c) are  $\delta$ -independent at the boundary level, and  $\delta$  is inert outright at the curve level (Fig. 6).

## E CONTROLLED-ADAPTER REPLICATION OF THE VERDICT MAP

The one-adapter limitation of §6 pre-registered a controlled  $2\times 2$  factorial: two plain-LoRA checkpoints trained under the original probe adapter’s frozen recipe on opposite style clusters (the top and bottom 12 artists by median latent redundancy, hereafter the high-redundancy, visually flat, and low-redundancy, heavily textured, clusters), with stem-level membership manifests frozen before training (124 training images each, identical seeds). Each adapter was probed on 48 stems spanning four membership cells ( $N=12$  each): trained-on, held-out images of trained artists, unseen artists of the same cluster, and unseen artists of the opposite cluster, with the cross-cluster stems shared between the two runs so cross-adapter contrasts read paired per stem. Figure 7 repeats the verdict-map recipe per route, overlaying both adapters.

Three reads, stated at the bin-mean level (per-cell tables reproduce from the archived per-image rows). First, the (route,  $\sigma$ ) shape replicates on both checkpoints (Fig. 7). Second, the factorial contrasts are null at instrument resolution: the probe-style main effect and the adapter $\times$ probe-style interaction (the pre-registered verdict quantity) are both bounded below the resolvable bin-mean effect (raw-gap paired interaction  $-0.022 \pm 0.027$  at 896,  $-0.015 \pm 0.059$  at 768), and the membership contrast (trained-on versus held-out) does not replicate in sign across the two adapters. The map is therefore adapter- and content-agnostic on this model at our resolution, supporting a calibrate-once-per-model reading. Third, the one quantity that is *not* checkpoint-invariant is the absolute redraw-floor level: in-window floor cosines sit at  $\approx 0.73$  for the high-redundancy-cluster adapter versus  $\approx 0.50$  for the low-redundancy-cluster adapter, uniformly across all four membership cells. The floor’s level is a property of the checkpoint (its gradient stochasticity), while the safety map built on top of it is not.

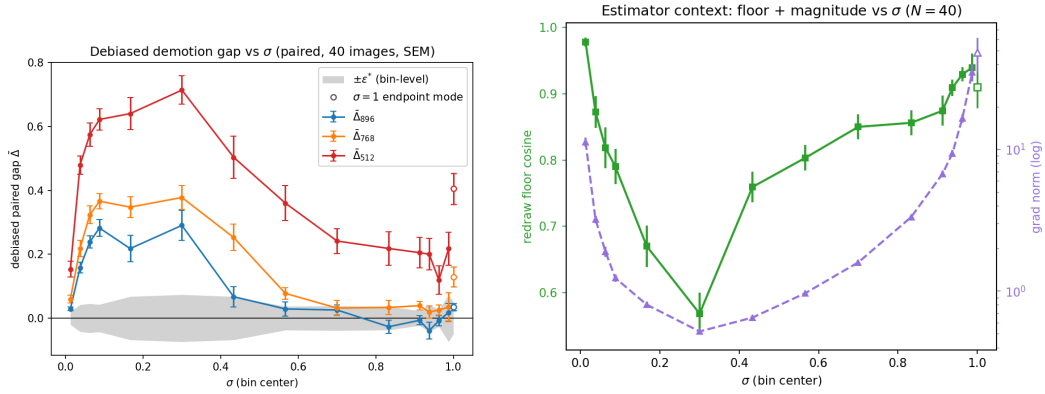


Figure 8: The measurement in detail. (a) Debiased demotion gap per  $\sigma$ -bin, all three routes on one shared axis, paired against the re-encoding control (1024-tier natives,  $N=40$ ; per-bin values and SEMs in Table 6; gray band: bin-level  $\pm\epsilon^*$  (§4.1); open markers: the  $\sigma=1$  endpoint bin); safety is a per-route boundary. The 896 route enters the band at  $\sigma \approx 0.5$ , 768 falls only to a low-excess shelf ( $+0.02$ – $0.04$  over  $0.65 < \sigma < 0.95$ , short of the strict margin), and 512 never falls inside the band (closest approach  $+0.12 \pm 0.05$  at  $\sigma = 0.96$ ); all three boundaries stand against the spectral family’s prediction of  $\sigma \approx 0.14$ – $0.20$ . (b) The estimator context those curves are read through: the redraw-floor cosine and the U-shaped gradient norm  $\|\bar{g}_{\text{src}}(\sigma)\|$ .

## F PER-BIN VERDICT NUMBERS, AND THE HISTORICAL RECORD

Table 6 is the numerical form of Fig. 8a, the paper’s verdict map, and Table 7 the debiased floor reads that §4.3 scores the graph term on. The remaining tables are the raw-estimator record: retained for continuity and for the reads defended in §4.1 (one-signed bias; paired same-grid contrasts), not cited as debiased evidence. Table 8 is the uncorrected counterpart of the verdict map on the original coarse grid, and its last two rows are the coarse-grid estimator context (Fig. 8b plots the same two curves on the dense verdict grid).

Table 6: **Debiased verdict map** (plotted in Fig. 8a): paired per-image debiased excess  $\overline{\text{gap}}$  (SEM) against the re-encoding control, 1024-tier natives ( $N=40$ ,  $D=12/\text{bin}$ , deterministic kernels, on a segmented grid dense below  $\sigma = 0.1$  and above  $0.9$ : 14 bins in  $(0, 1)$  plus a  $\sigma=1$  endpoint bin; trimmed per §4.1, retaining 36–40 images per cell). Bin-level  $\epsilon^*$  at this  $(N, D)$  is  $0.02$ – $0.07$ . A reconstruction check (the realized excess rebuilt from per-arm means) flags the three  $\sigma \leq 0.06$  bins as partly estimator inflation, so the raw-paired excess is the primary read there (896:  $+0.037/+0.118/+0.221$ ; 768:  $+0.065/+0.202/+0.275$ ; 512:  $+0.172/+0.406/+0.497$ ), up to  $0.08$  smaller with the same verdicts.

| $\sigma$ bin center | 1024→896      | 1024→768      | 1024→512      |
|---------------------|---------------|---------------|---------------|
| 0.013               | +0.029 (.006) | +0.059 (.012) | +0.154 (.025) |
| 0.038               | +0.158 (.016) | +0.219 (.026) | +0.480 (.029) |
| 0.063               | +0.239 (.019) | +0.323 (.028) | +0.575 (.037) |
| 0.088               | +0.282 (.026) | +0.366 (.024) | +0.622 (.033) |
| 0.167               | +0.218 (.041) | +0.348 (.033) | +0.640 (.050) |
| 0.300               | +0.291 (.047) | +0.378 (.038) | +0.714 (.045) |
| 0.433               | +0.067 (.032) | +0.253 (.041) | +0.504 (.065) |
| 0.567               | +0.029 (.023) | +0.078 (.019) | +0.360 (.055) |
| 0.700               | +0.026 (.016) | +0.032 (.023) | +0.242 (.039) |
| 0.833               | −0.027 (.020) | +0.034 (.023) | +0.218 (.053) |
| 0.913               | −0.007 (.014) | +0.039 (.013) | +0.205 (.047) |
| 0.938               | −0.039 (.026) | +0.019 (.019) | +0.200 (.049) |
| 0.963               | −0.008 (.015) | +0.026 (.016) | +0.120 (.045) |
| 0.988               | +0.018 (.028) | +0.034 (.045) | +0.217 (.052) |
| 1.0 (endpoint)      | +0.034 (.011) | +0.130 (.030) | +0.406 (.049) |

Table 7: **The floor, debiased** (scored in §4.3). **Endpoint:** at  $\sigma=1$  the input is exactly  $\epsilon$  on every grid, so the input-mediated data term vanishes by construction. **x-zero:** the image is removed from input *and* target; any surviving gap is pure graph-shape sensitivity. Debiased draw-limit values  $\widehat{\text{gap}}_\infty$  [bootstrap 95% CI over images] from nested draw sweeps (endpoint:  $N=12$ ,  $D = 4 \dots 64$ ; x-zero:  $N=40$ ,  $D = 4 \dots 32$ ). The two columns agree within CI at every route; our pre-registered falsification criterion (all endpoint gaps  $\approx 0$ ) is not met; and no member of the spectral family can produce a nonzero entry here.

| route                      | endpoint $\widehat{\text{gap}}_\infty$ [95% CI] | x-zero $\widehat{\text{gap}}_\infty$ [95% CI] |
|----------------------------|-------------------------------------------------|-----------------------------------------------|
| native self-floor (cosine) | 1.005 [0.994, 1.016]                            | —                                             |
| re-encoding control        | −0.003 [−0.017, +0.008]                         | —                                             |
| 1024→896                   | +0.019 [+0.010, +0.030]                         | +0.034 [+0.017, +0.058]                       |
| 1024→768                   | +0.056 [+0.043, +0.071]                         | +0.074 [+0.053, +0.094]                       |
| 1024→512                   | +0.304 [+0.197, +0.424]                         | +0.283 [+0.232, +0.332]                       |

Table 8: **Raw estimator** (historical record; superseded by Table 6): demotion gap by  $\sigma$ -bin, 1024-tier natives ( $N=40$ , bin-mean SEM  $\sim 0.02$ , re-encoding control  $|\text{gap}_{\text{reenc}}| \leq 0.054$  everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. Raw values understate mid- $\sigma$  gaps (the floor attenuates the read) and overstate endpoint floors (single-estimate variance bias); the debiased map corrects both.

| $\sigma$ bin center              | 0.06 | 0.19 | 0.31 | 0.44 | 0.56        | 0.69        | 0.81        | 0.94 |
|----------------------------------|------|------|------|------|-------------|-------------|-------------|------|
| gap <sub>896</sub> (ratio 0.875) | .110 | .162 | .148 | .137 | <b>.048</b> | <b>.030</b> | <b>.030</b> | .053 |
| gap <sub>768</sub> (ratio 0.75)  | .144 | .216 | .208 | .223 | .164        | .163        | .115        | .063 |
| gap <sub>512</sub> (ratio 0.5)   | .348 | .410 | .355 | .469 | .430        | .391        | .289        | .296 |
| cos <sub>floor</sub>             | .84  | .65  | .51  | .65  | .70         | .77         | .80         | .83  |
| $\ g\ $ (native)                 | 2.6  | 0.5  | 0.4  | 0.5  | 0.7         | 1.1         | 2.0         | 9.3  |

Table 9: Iso-severity probe (**raw estimator**): gap by  $\sigma$ -bin on the 1280-tier cache ( $N=24$ , 4 draws/bin) against the corpus 1024→896 curve. The ratio-matched routes coincide within  $\sim 1.4$  combined SEM at every bin despite a  $1.6\times$  difference in absolute target size; the same-native harsher-ratio route separates. \*896’s canonical 16-draw endpoint is −0.009 raw, +0.019 debiased (Table 7).

| $\sigma$               | 0.125 | 0.375 | 0.625 | 0.875 | 1.0   |
|------------------------|-------|-------|-------|-------|-------|
| re-encoding control    | +.047 | −.005 | +.048 | +.020 | −.012 |
| 1280→1120 (ratio .875) | .129  | .110  | .054  | −.012 | −.007 |
| 1280→1024 (ratio .80)  | .170  | .178  | .111  | .002  | .061  |
| 1024→896 (ratio .875)  | .132  | .076  | .077  | .049  | .100* |

Table 10: Ratio-transfer run (**raw estimator**): 896-tier natives ( $N=40$ ), pre-registered bar “gap<sub>768</sub> within the re-encoding band at  $\sigma \geq 0.5$ ”: **fails** (residual 0.06–0.12,  $\sim 2\times$  the 1024→896 plateau;  $\sim$ half of it is expected estimator bias, §4.1; the separation is independently confirmed by the debiased same-grid floors, §4.3).

| $\sigma$ bin | 0.06 | 0.19 | 0.31 | 0.44 | 0.56 | 0.69 | 0.81 | 0.94 |
|--------------|------|------|------|------|------|------|------|------|
| 896→768      | .114 | .209 | .168 | .143 | .124 | .056 | .092 | .061 |
| 896→512      | .318 | .372 | .308 | .340 | .320 | .280 | .329 | .217 |

Table 11: Route-uniformity (in amplitude) of the mean prediction residual: cross-grid excess of  $\bar{r} = \mathbb{E}_\epsilon [\hat{v} - (\epsilon - x)]$  over split-half floors (relative  $L_2$ ; same-grid re-encoding control  $\leq 0.02$  everywhere). The three main routes coincide within  $\sim \pm 0.02$  at every  $\sigma$  while their gradient floors span 0 to 0.3. This is a norm read; the direction-resolved follow-up (Appendix B) shows the underlying mismatch directions are image- and route-specific.

| $\sigma$  | 0.125 | 0.375 | 0.625 | 0.875 | 1.0  |
|-----------|-------|-------|-------|-------|------|
| 1280→1024 | .885  | .825  | .715  | .465  | .360 |
| 1024→896  | .862  | .808  | .706  | .459  | .378 |
| 896→768   | .860  | .814  | .715  | .466  | .393 |
| 768→512   | .897  | .872  | .767  | .508  | .435 |

Table 12: Depth localization of the floor (**raw estimator**): mean per-block gap at  $\sigma=1$ , early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry  $\sim 3\times$  the late-block gap. The apparent content share (ep – xz) is a late-block minority effect in raw units; the debiased draw-limit comparison finds ep = xz within CI at the whole-adapter level (Table 7). Within a block every module type sits in 0.22–0.31 at 512/ $\sigma=1$ , including the query and key rows of the fused QKV projection: landing-side uniformity, which is why the positional share needed an origin-side intervention to find.

| route    | ep early | ep late | xz early | xz late |
|----------|----------|---------|----------|---------|
| 1024→512 | .357     | .223    | .351     | .125    |
| 1024→768 | .164     | .085    | .121     | .033    |
| 1024→896 | .023     | .015    | .044     | .012    |

Table 13: Positional-interpolation intervention at the  $\sigma=1$  endpoint ( $N=40$ , 16 draws; **raw** arm values; the paired  $\Delta$  column is a same-grid contrast in which the finite-draw bias cancels to first order): exact phase-geometry alignment erases the mild-route floor and  $\sim 30\%$  of the harsh-route floor.

| route              | plain (SEM)   | PI-aligned (SEM) | paired $\Delta$ (SEM) | improved |
|--------------------|---------------|------------------|-----------------------|----------|
| 1024→896 (control) | −0.021 (.041) | −0.040 (.040)    | —                     | —        |
| 1024→768           | +0.080 (.048) | −0.001 (.039)    | +0.081 (.031)         | 78%      |
| 1024→512           | +0.320 (.058) | +0.224 (.056)    | +0.096 (.039)         | 70%      |